Mission-Enabling Nuclear Electric Propulsion Revisited: an audited, reconciled and optimal-control reanalysis of the SPC 536 study

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Abstract

Nuclear electric propulsion (NEP) removes the power limit that constrains electric propulsion far from the Sun. Our 2025 course paper argued that it is therefore mission enabling, including a 5 t Mars cargo delivered in 200 days by a 100 kWe reactor. An audit of that paper found 44 errors, and this paper rebuilds the analysis. The original calculation was first reproduced and then corrected: the NEP transfer time is 3.00 yr, not 0.85, and the 200-day mission cannot be closed. A trajectory simulation and a SMAD-style subsystem study, which disagreed, were then run on one documented input set together with a new optimal-control solver and a seeded Monte Carlo. The simulation's technology-tier Mars results were found to come from failed trajectory-table nodes. Once these are removed, the two models agree to 5–15 % for Mars (8 t in 450 d needs 95, 64 and 80 kWe in the 2020s, 2030s and 2040s), and for the outer planets they differ only through the choice of capture definition. Burn-coast-burn trajectories are upper bounds, with trip times 3–46 % longer than the optimum at equal propellant. The mission verdicts are set by technology readiness rather than by mass closure, and the converter life test is the critical path of both future eras. Throughout, results marked *computed* are model outputs, *literature* values are cited, and the 2030s and 2040s technology states are *assumed* milestones, not forecasts.

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Nomenclature

Propulsion Parameters

F	thrust [N]
$\dot{m}$	propellant mass flow rate [kg/s]
v_e	effective exhaust velocity [m/s]
I_{sp}	specific impulse [s]
g_0	standard gravitational acceleration [m/s^2]
η_t, η_{ppu}	thruster efficiency and power processing unit (PPU) efficiency
P_{thr}, P_{jet}	electric power to the thrusters and jet power [W]
a_0	initial thrust acceleration, F/m_0 [m/s^2]

Power System Parameters

P_e, P_{bus}	electrical power output and bus load [W]
f_{EP}	fraction of electrical power allocated to propulsion
α	power-system specific mass [kg/kWe] (in Eq. (8), the steering angle [rad])
τ, λ	RTG decay time constant [yr] and NEP degradation rate [yr^{-1}]
P_{waste}, Q_{rej}	waste heat to be rejected [W]
A_{rad}, T_r, T_s	radiator area [m^2], radiator temperature and sink temperature [K]
ε, σ, F_{eff}	emissivity, Stefan–Boltzmann constant, effective view factor

Mission Parameters

ΔV	velocity change required for the mission [km/s]
C_3	launch energy, the square of the hyperbolic excess speed [km^2/s^2]
m_0	initial spacecraft mass at Earth escape [kg]
m_p, m_{pl}, m_{bus}, m_{ps}, m_{thr}	propellant, payload, bus, power-system and thruster masses [kg]
c, f_{tank}	unallocated reserve and tank mass fraction
T, t_{burn}, t_{trip}	transfer time, thrusting time and trip time [d or yr]
P_{close}	probability that a mission closes in the Monte Carlo

1 Introduction

Electric propulsion offers an order of magnitude more specific impulse than chemical propulsion, but beyond the inner solar system it is limited by the power that solar arrays or radioisotope generators can supply rather than by thruster physics. Nuclear electric propulsion (NEP) replaces these sources with a fission reactor whose output does not fall with distance from the Sun. The 2025 course paper [1] asked whether NEP is therefore "mission enabling". Its headline claims were that NEP transfers are about twenty times faster than solar-electric ones, that a 100 kWe NEP system makes a 5 t, 200-day Mars cargo mission possible, and that NEP has the smallest specific mass of the power sources compared. A line-by-line audit of that paper (44 findings: 7 critical, 18 major, 13 minor, 6 typographical; `audit/audit_report.md`) found that the qualitative argument partly survives, since a decaying RTG cannot power a fast mission and NEP power does not fall with distance. The quantitative conclusions, however, do not survive. The errors that change a conclusion are:

- **Table 1 is inconsistent with the paper's own Eq. (26).** The NEP time printed as 0.85 yr is 3.16 yr from the equation (ratio 0.27), and the solar and RTG columns are all scaled by an unexplained factor of about 0.53. Because time is inversely proportional to power for a fixed thruster, the "20× faster" claim is therefore 10× at best.
- **The constant-mass propellant law** (Eq. 19/26) gives a resistojet propellant load of 207 % of the spacecraft mass, so the "73-day Mars transfer" is physically impossible.
- **The Mars case study (Table 3) mixes alpha conventions** (per kWe versus per kW of thruster power) and omits the thrusters, PPU, tanks and bus. It concludes that "solar is prohibitive" when its own table shows solar needing only 17 % more power and 18 % more initial mass. The 200-day transit is also shorter than the 259-day Hohmann flight time, and the empirical $\Delta V(T)$ law (Eq. 27) falls below the Hohmann minimum for $T > 220$ d.

- **The RTG decay constant** (75 yr) is not that of the MMRTG, which is 40.1 yr (110 We falling to 72 We in 17 yr [2]). With the paper's own decay law, a ten-RTG craft can never deliver the energy it needs (ratio 0.35).
- **The abstract's "smallest specific mass"** is contradicted by the paper's own solar value, and no outer-planet mission is ever computed, although the argument for NEP rests on distance.

There therefore remains a gap between what the original study claims and what its own equations and inputs support, and no consistent quantification of where NEP is in fact enabling. The goal of this paper is to address this gap. This study rebuilds the analysis in four steps and keeps every number traceable to a model output or a source.

The rest of the paper is structured as follows. Section 2 sets out the equations, inputs and verdict rule, and Section 3 validates the models. In the Results (Section 4), the original calculation is first reproduced exactly and then corrected (Sections 4.1–4.2). Two independently built models, a trajectory-integrating simulation (`sim`) and a SMAD-style subsystem design study (`smad`), which disagreed in places, are then reconciled on a single documented input set, and their trajectory sub-models are checked against a new optimal-control solver (Section 4.3). Next, a feasible redesign and a thermal and subsystem analysis are presented for three technology eras (Sections 4.4–4.6), followed by the mission verdicts and probabilities of closure (Section 4.7). The design ideas that survive a paired Monte Carlo test and a literature check are then listed with their gains and precedents (Section 4.8), together with the sensitivities (Section 4.9). The Discussion (Section 5) covers the limitations and the strongest objections a reviewer would raise, and the paper concludes with an assessment of where NEP is and is not enabling (Section 6). Throughout, we mark which results are computed by the models, which are literature values and which depend on speculative technology assumptions: *computed* (model output), *literature* (a value read in a source), *assumed* (a stated assumption with a range).

2 Approach

The goal of this study is to determine, with consistent inputs, when replacing solar arrays or radioisotope thermoelectric generators (RTGs) with a fission power system makes an electric-propulsion mission possible. The analysis links

1. the power source (solar, RTG or NEP) and its thermal sizing,
2. the electric thrusters and the mass closure of the vehicle,
3. a heliocentric trajectory model with a planetocentric capture, and
4. the launch capacity, an uncertainty model and a verdict rule,

to compute power, mass, trip time and probability of closure for each mission and technology era. The equations and inputs used for each part are given below.

2.1 Propulsion and mass closure

Thrust and mass flow follow from the jet-power relation $P_{jet} = \frac{1}{2} F v_e = \eta_t P_{thr}$ with $v_e = g_0 I_{sp}$ [3]:

$$F = \frac{2\,\eta_t\,\eta_{ppu}\,f_{EP}\,P_e}{g_0 I_{sp}}, \qquad \dot{m} = \frac{F}{g_0 I_{sp}}, \tag{1}$$

$$f_{EP} = 1 - \frac{P_{bus}}{P_e}, \qquad m_p = m_0 \left(1 - e^{-\Delta V / v_e}\right), \tag{2}$$

$$t_{burn} = \frac{m_p\, v_e^2}{2\,\eta_t\, P_{thr}} \quad (P_{thr}, I_{sp} \text{ constant}). \tag{3}$$

where F is the thrust, $\dot{m}$ the mass flow, P_e the electrical power, m_p the propellant mass and t_{burn} the thrusting time. The original paper's Eq. (26) is the same expression with $m_p \to m_0 \Delta V / v_e$ (constant mass), which overstates the propellant by up to a factor of two. We keep the PPU and thruster efficiencies separate (η_{ppu} is 0.90 for NSTAR [4] and 0.93 in the SMAD study). The generic thruster efficiency is $\eta_t(I_{sp}) = \eta_{max}/(1 + (I_{sc}/I_{sp})^2)$ with $\eta_{max} = 0.763$ and $I_{sc} = 1195$ s; it was fitted to six verified operating points and is used only between 1800 and 4300 s. The mass of the vehicle at Earth escape then closes as

$$m_0 = m_{pl} + (1 + c)\,(m_{bus} + m_{ps}(P_e) + m_{thr}) + (1 + f_{tank})\,m_p, \tag{4}$$

where c is the unallocated reserve (0.10, the SMAD system margin), $f_{tank} = 0.10$ [5], $m_{thr} = 3.57$ kg/kW of thruster rating plus a PPU of 5 kg/kW, and the power-system mass m_{ps} is read from the tier tables of the

SMAD study (Section 4.6). The thruster count is taken as the larger of $\lceil P_{thr}/P_{unit} \rceil$ and the number that keeps the hours and xenon throughput per unit below 80 % of the demonstrated values (NEXT: 51,184 h, 918 kg [6]; NSTAR: 30,352 h, 235 kg [7]).

2.2 Power sources and thermal sizing

Solar array power is modelled with an inverse-square law in distance and an exponential degradation in time, RTG power with an exponential decay, and NEP power with a slow exponential degradation from fuel burnup and component ageing. The waste heat of the fission system sets the radiator area:

$$P_{solar}(r,t) = P_{1\,AU}\left(\tfrac{1\,\mathrm{AU}}{r}\right)^2 e^{-\lambda_s t}, \qquad P_{RTG}(t) = P_0\, e^{-t/\tau}, \quad \tau = \frac{17\ \mathrm{yr}}{\ln(110/72)} = 40.1\ \mathrm{yr}, \tag{5}$$

$$P_{NEP}(t) = P_0\, e^{-\lambda t}, \quad \lambda = -\ln(0.95)/15\ \mathrm{yr} = 3.42\times10^{-3}\ \mathrm{yr}^{-1}, \qquad P_{waste} = P_e\left(\tfrac{1}{\eta_{sys}} - 1\right), \quad A_{rad} = \frac{P_{waste}}{\varepsilon\sigma F_{eff}\,(T_{rad}^4 - T_s^4} \tag{6}$$

where τ is the RTG decay constant, λ the NEP degradation rate and T_{rad} and T_{sink} the radiator and sink temperatures. It follows that an RTG can deliver at most $P_0\tau$ of energy over its life. The radiator view factor $F_{eff} = 1.50$ is calibrated to the 26.9 m^2 of the El-Genk–Tournier design [8]; in the SMAD study the area also includes a 10 % margin and a fin efficiency (Section 4.5).

2.3 Trajectory models

All trajectory models treat planar heliocentric two-body motion with thrust from Eq. (1). Departure is at Earth escape, with hyperbolic excess speed $\sqrt{C_3}$ along Earth's velocity. Arrival is defined as rendezvous with the *orbit* of the target planet (circular, coplanar, semi-major axis from [9]) with free phase, so launch-window waiting is not costed. It is followed by a tangential capture spiral from rest to a circular orbit of radius r_c (20 planetary radii for Mars, 30 for the giants):

$$\Delta V_{cap} = \sqrt{\mu_p/r_c}, \qquad t_{cap} = \frac{m_{p,cap}\, v_e^2}{2\eta_t P_{thr}}. \tag{7}$$

Three heliocentric models are compared in Section 4.3.

(a) Burn-coast-burn family (sim). This model flies a velocity-aligned burn, a Kepler coast and a second velocity-aligned burn, and is solved by a one-dimensional shooting in the junction semi-major axis. The resulting trajectory is feasible but not optimal.

(b) The SMAD $\Delta V(T)$ law. The SMAD study uses $\Delta V_{EP}(T, C_3) = \sqrt{\Delta V_\infty^2 + \beta^2/T^2} + \sqrt{C_{3,ref}} - \sqrt{C_3}$, with $\beta^2 = (\Delta V_{ref}^2 - \Delta V_\infty^2)T_{ref}^2$ calibrated at one point per case, and $t_{trip} = t_{burn}/\mathrm{duty}$. The law includes the capture, but it is a heuristic.

(c) Optimal control (new). The new solver uses the state (r, θ, v_r, v_t, m) in canonical units (1 AU, $\sqrt{\mathrm{AU}^3/GM_\odot}$), with the steering angle α from the local horizontal and the throttle $u \in [0,1]$ as controls:

$$\dot r = v_r, \qquad \dot\theta = v_t/r, \qquad \dot v_r = \frac{v_t^2}{r} - \frac{1}{r^2} + \frac{uF}{m}\sin\alpha, \quad \dot v_t = -\frac{v_r v_t}{r} + \frac{uF}{m}\cos\alpha, \quad \dot m = - \tag{8}$$

$$F = \frac{2\eta_t P_{thr0}}{v_e}\left(\tfrac{1\,\mathrm{AU}}{r}\right)^n e^{-\lambda t}, \qquad n = 0\ \text{(fission)},\ n = 2\ \text{(solar)}.$$

The boundary conditions are $r(0) = 1$, $v_t(0) = v_c + \sqrt{C_3}$, $v_r(0) = 0$ and $r(T) = r_t$, $v_r(T) = 0$, $v_t(T) = \sqrt{1/r_t}$. Power-limited thrust at constant I_{sp} is bang-bang, so the *minimum-time* problem uses $u \equiv 1$ and the *minimum-propellant* problem at fixed $T \geq T_{min}$ lets u drop to zero on coast arcs. The controls are piecewise linear in normalised time (14–20 steering nodes, 16–24 throttle nodes) and are integrated with fixed-step RK4 (400–500 steps, numba). The nonlinear program is solved with SLSQP from 45 steering starts, followed by a refinement with 20 nodes. A solution found this way is a *feasible* trajectory and therefore an *upper bound* on the true minimum time and propellant; it is not a proof of global optimality (Section 3). For mass closure, the solutions are tabulated against the initial acceleration $a_0 = F/m_0$, the exhaust velocity and the time ratio $s = T/T_{min}$ (9×3×6 nodes per target and C_3) and then interpolated. This is exact for a distance-independent source, because the heliocentric motion then depends only on a_0 and v_e.

Table 1: Selected inputs read from `data/params.json` and `sim/extra_params.json`; the full list with status for every key is in `sim/results/parameters_used.json`. Source ids refer to `data/sources.md`.

Input	Value	Source id	Status
NSTAR Isp, total efficiency	3270 s / 0.65	[5]	VERIFIED
NSTAR PPU efficiency	0.9	[5]	VERIFIED
NSTAR demonstrated life / throughput	30352 h / 235 kg	[6]	VERIFIED
NEXT Isp, total efficiency	4170 s / 0.706	[7]	VERIFIED
NEXT demonstrated life / throughput	51184 h / 918 kg	[8]	VERIFIED
MMRTG power after 17 yr (BOL 110 We)	72 We	[16]	VERIFIED
MMRTG specific mass	409 kg/kWe	[16] derived	DERIVED
Dawn array power at 1 AU	10.3 kW	[30]	VERIFIED
solar array degradation	0.0075 /yr	[1] (terrestrial basis)	UNVERIFIED
solar array specific mass	[25, 40] kg/kW	[37]	VERIFIED
fission power decay rate	0.00342 /yr	[2] design goal	DERIVED
housekeeping bus load (sim)	5 kWe	assumption	UNVERIFIED
capture orbit, planets / giants	20 R_eq / 30 R_eq	assumption	UNVERIFIED
largest radiator studied	2500 m^2	[21]	VERIFIED
propellant tank mass fraction	0.1 tank mass / propellant mass	SMAD Table 10-12	UNVERIFIED
launch-vehicle margin	0.1	assumption	UNVERIFIED

2.4 Launch, uncertainty and verdict rule

Launch capacity is modelled as $m(C_3) = m_{LEO} \exp(-\Delta V(C_3)/v_{e,eff})$, with $\Delta V(C_3) = \sqrt{C_3 + 2\mu/r} - \sqrt{\mu/r}$ at $r = 6778$ km. The pair $(m_{LEO}, v_{e,eff})$ is (63.8 t, 2.39 km/s) for an expendable Falcon Heavy (fitted to 10.9 t at $C_3 \approx 24$ [10]), (105 t, 3.2 km/s) for SLS Block 1B (availability uncertain) and (100 t, 4.0 km/s) for a refuelled Starship (a pure assumption). A 10 % vehicle margin is included in the capacity, and a further 1.15 design reserve is applied. Closure probabilities are computed with a Latin hypercube Monte Carlo ($N = 2000$, seed 20251103) over the power-system mass multiplier, I_{sp} (±5 %), thruster efficiency (0.90–1.02), PPU efficiency (0.84–0.94), thruster life (×0.5–1.5; 0.25–2 for the 2040s), launch capacity (×0.85–1.05), bus mass (×0.9–1.25) and degradation (×1–3), with the hardware fixed and the trajectory re-planned. A mission *closes* in a sample when the propellant fits, the trip time is within the limit, the thruster life and throughput hold, the stack fits the vehicle and the radiator is within the largest studied (2,500 m^2 [11]). The verdict combines this physical closure with the readiness of the technology tier through the SMAD rule: POSSIBLE if $P_{close} \geq 0.8$, the minimum critical TRL is ≥ 6 and the converter life requirement is $\leq 1\times$ the demonstrated hours; MARGINAL if $P_{close} \geq 0.5$, TRL ≥ 4 and life $\leq 2\times$; otherwise NOT POSSIBLE. These thresholds are our judgement, not a standard.

2.5 Inputs, sources and status

Table 1 lists the principal inputs. Every number in the models is read through a parameter store that records its source and a status: VERIFIED means a document was opened and the value seen, DERIVED means computed from verified values, and UNVERIFIED means secondary, search-snippet only or an assumption. Of the 148 keys read by the simulation, the UNVERIFIED ones are marked in `sim/results/parameters_used.json`, and the tier assumptions of the SMAD study are kept in its assumption register (`smad/results/assumption_register.csv`).

2.6 Key assumptions

Finally, we summarise the key modelling assumptions, each of which is discussed again in Section 5:

- Each thruster operates at a fixed I_{sp} and efficiency; no throttle curves are modelled.
- Planets are on circular, coplanar orbits with free phase; launch-window waiting, inclination and Earth escape from low orbit are not costed.
- Optimal-control solutions are feasible trajectories and are treated as upper bounds on the true minimum time and propellant.
- Power-system masses are taken from the conceptual tier tables of the SMAD study; the launch curves for SLS Block 1B and a refuelled Starship are partly or wholly assumed.
- The 2030s and 2040s technology tiers are *assumed* states conditional on stated milestones, not forecasts, and the verdict thresholds are our judgement.

Table 2: Simulation validation: largest relative error by family (59 checks, all within tolerance).

Family	Check	Max rel. error
V1 tangential	integrated tangential-spiral ΔV vs $\lvert v_1 - v_2 \rvert$	0.0029
V2 angular-momentum	angular momentum, 5 orbits	6.3e-12
V2 energy	energy, 5 orbits	1.7e-11
V2 position	position, 5 periods	8e-10
V3 Hohmann	Hohmann ΔV vs params	0.00011
V3 integrated	integrated Hohmann apoapsis	3.9e-12
V4 integrated	Tsiolkovsky, integrated	3.9e-15
V4 propellant	propellant fraction	5e-16
V5 RK4	RK4 vs analytic	8e-15
V5 spiral	analytic radius-from-ΔV model	0.018
V6 table	trajectory-table interpolation	0.026
V7 high-thrust	burn-coast-burn $\rightarrow$ Hohmann	0.0058
V7 low-thrust	burn-coast-burn $\rightarrow$ spiral	0.00033
V8 AMTEC	AMTEC preset vs El-Genk	9.5e-05
V8 Brayton	Brayton preset vs Mason	2.2e-16
V8 MMRTG	MMRTG 72 We at 17 yr	0
V8 scaling	power-system scaling law	0.092
V8 solar	solar array law	1.9e-16
V8 thermal	radiator equation	1.3e-16
V8 thrust-power	thrust-power law vs NSTAR 92 mN	0.00026
V9 Dawn:	Dawn-like benchmark (thrust hours)	0.22

3 Model Validation

Before the models are used for any mission result, each is checked against analytic solutions, flight data or published designs. This section summarises those checks and states what they do not cover.

Simulation (sim). The simulation passes 59 of 59 automated checks (Table 2). The ballistic energy and angular-momentum drift is of order 10^{-11} over five orbits, the integrated Hohmann ΔV is within 10^{-4} of the tabulated values, the Tsiolkovsky equation holds to machine precision, and the thrust-power law is within 0.03 % of the measured NSTAR thrust. The power-system presets reproduce the El-Genk 3390 kg / 26.9 m^2 design and the Mason 4115 kg Brayton design. Finally, a Dawn-like benchmark (a planar spiral from 1 AU to Ceres with the NSTAR throttle table) returns 385 kg of xenon against the 416 kg consumed by Dawn [12], a ΔV of 11.6 against 11.5 km/s, and 39,700 against 51,250 thrust hours. The 22 % deficit in hours comes from the assumed throttle table and an unverified housekeeping load, and the match in ΔV is partly fortuitous, because Dawn's Mars flyby and the omissions of the planar model offset each other.

Limits of that validation. The check suite covers the trajectory table (V6, 2.6 % interpolation error) only at the nodes that the integrator solved. It did not test whether every node was solved. Section 4.3 shows that, for two of the tables used in the technology-tier study, 35 of 63 and 7 of 63 nodes had failed and had been filled with neighbouring values, an error that the suite could not detect.

SMAD study. The power-system, shield, thermal and mass models of the SMAD study are validated against the JIMO Brayton design (the shield model is conservative by about a factor of 1.9 on mass), against the NASA 20 kWe Uranus mass-equipment list [10] and against Mason's 20/100/200 kWe specific-mass curve [13], with differences of −28 %, +6 % and +25 % respectively. Above about 300 kWe the model flattens at 40 kg/kWe, whereas the literature keeps falling (13 kg/kWe at 1.9 MWe [14]), so it must not be used in that range.

Optimal-control solver (new). Table 3 and the checks below test the solver itself. (i) With zero thrust, the integrator holds a circular orbit for five periods to $2e-12$ in radius and reproduces a Hohmann half-ellipse to 2e-14 in apoapsis radius (flight time 259 d, the analytic 259-d value). (ii) The optimal controls, re-integrated with an independent adaptive integrator (DOP853, relative tolerance 10^{-11}), miss the target orbit by at most 5e-06 in canonical units; the fixed-step RK4 shooting is therefore not an artefact. (iii) In the convergence test on the number of steering nodes and steps, the minimum times differ by 0.10 % between 14 and 28 nodes. (iv) In the low-thrust limit ($a_0 \rightarrow 0$), the flown ΔV approaches the Edelbaum value $\lvert v_{c1} - v_{c2} \rvert$ to within 2.5 %. (v) With a time budget of twice the minimum time, the propellant fraction tends to 0.161, against 0.160 for the impulsive Hohmann transfer. (vi) Every optimal solution found is a feasible trajectory, so $T_{opt} \leq T_{bcb}$ must hold wherever the burn-coast-burn solution is valid; this held in all comparisons of Section 4.3. Two things are *not* validated. The first is global optimality: a multi-start search

Table 3: Checks of the optimal-control solver (`reconcile/oc_validate.py`).

Optimal-control solver check	Value
Coast, eccentric orbit, 5 periods: energy drift	2.4e-12
Hohmann coast, apoapsis radius error	2.4e-14
Low-thrust limit, a_0 = 3e-05 m/s^2: ΔV / Edelbaum −1	+0.025
Mars, N=300, K=10: T_{min} = 265.6 d, DOP853 terminal error	1.1e-07
Mars, N=400, K=14: T_{min} = 265.2 d, DOP853 terminal error	1.2e-06
Mars, N=500, K=20: T_{min} = 265.1 d, DOP853 terminal error	5.2e-06
Mars, N=700, K=28: T_{min} = 264.9 d, DOP853 terminal error	8.3e-07
Jupiter, N=300, K=10: T_{min} = 768.6 d, DOP853 terminal error	3.9e-08
Jupiter, N=400, K=14: T_{min} = 768.5 d, DOP853 terminal error	1.9e-06
Jupiter, N=500, K=20: T_{min} = 768.2 d, DOP853 terminal error	7.3e-07
Jupiter, N=700, K=28: T_{min} = 768.2 d, DOP853 terminal error	1.3e-06
Propellant fraction at 2 T_{min} vs impulsive Hohmann (0.1601)	0.1606

Table 4: Case A: Table 1 of the paper re-computed from its own Eq. (26); last column is the corrected model (rocket equation, PPU 0.90, decaying RTG; "never" = the source cannot deliver the energy).

Row (10 km/s, 20 t, NSTAR)	Paper [yr]	Eq. 26 [yr]	Paper / Eq. 26	Corrected [yr]
Single MMRTG	1900	3588	0.53	never
Cassini (3 RTGs)	240	446	0.54	never
Scaled (10 RTGs)	190	359	0.53	never
Solar (Dawn) @1 AU	17	32	0.54	33.79
Solar @5 AU	420	789	0.53	never
NEP 100 kWe	0.85	3.16	0.27	3.00

is used, and 12 of 18 starts agree on the same optimum in the test case, but there is no indirect-method certificate. The second is the planetocentric capture, which is analytic. The optimal-control times should therefore be treated as upper bounds on the true minima.

4 Results

This section applies the framework of Section 2 in the order in which the study was carried out. The original calculation is first reproduced (Case A) and corrected (Case B); the two independent models are then reconciled, a feasible redesign and its thermal and subsystem sizing are presented, and the mission verdicts, the design ideas that survive testing and the sensitivities follow.

4.1 Reproducing the original study (Case A)

We first re-implement the paper exactly as written (the Eq. 26 constant-mass law, no PPU, the Eq. 27 empirical $\Delta V(T)$ and Eq. 28) with its stated inputs, in order to find where each printed number comes from.

Table 1. Table 4 shows that none of the printed times follows from Eq. (26). Five rows are low by a constant factor of 0.53–0.54, and the NEP row by 0.27. Corrected with the rocket equation and a PPU efficiency of 0.90, a 100 kWe NEP craft needs 3.00 yr, not 0.85 yr, and the RTG rows have no solution at all, because the source cannot deliver the energy.

Table 2. The NEP column reproduces exactly from $t = m_0 \Delta V g_0 I_{sp}/(2\eta P)$ (692 d for NSTAR at 50 kW, 422 d for the SPT-100, and so on). The solar column reproduces only if the mean solar power is 3.16 kW rather than the stated 2.2 kW, and the RTG columns cannot be produced from any single power. With the constant-mass law, the resistojet case needs 41,480 kg of propellant for a 20 t craft and the arcjet case 20,394 kg (102 %), which no spacecraft can carry.

Table 3 and Fig. 8. The arithmetic of the Mars cargo table is correct given its inputs (Table 5); the problem lies in the inputs themselves. The paper's own Eqs. (27)–(28) place the infinite-power asymptotes at 112 d (NEP) and 130 d (solar), not at the 150 and 250 d of its Fig. 8, and Eq. (27) returns 2.5 km/s at 500 d, which is below the 5.59 km/s heliocentric Hohmann minimum. Figure 1 redraws the paper's Figs. 4, 6, 7 and 8 with these corrections.

Table 5: Case A reproduces the paper's Table 3 to its printed precision (rows 1–2). Case B applies the audit corrections to the same mission definition (AMTEC power-system model, PPU, thrusters, tanks, 1 t bus): the required electric power is 250 kWe, 27% payload fraction, and the 200-d definition is shorter than the 259-d Hohmann time.

Mars cargo, 5 t, 200 d	P paper [kW]	P recomputed [kW]	m_{ps} paper [kg]	m_{ps} rec. [kg]	m_0 paper [t]	m_0 rec. [t]
NEP	58	57.7	1904	1904	9.5	9.50
Solar	68	68.1	3146	3145	11.2	11.20
NEP, corrected (Case B)	-	250	-	6008	-	18.5
Solar, corrected (Case B)	-	none	-	-	-	-

Table 6: Principal Corrections from Case A to Case B (full list: `sim/changes_log.md`).

Item	Paper	Corrected	Source
Propellant law	constant mass, $m_0 \Delta V / v_e$	$m_0(1 - e^{-\Delta V / v_e})$	audit F5
PPU	none	$\eta_{ppu} = 0.90$	NSTAR [4]
MR-510 arcjet η	0.45 (inferred)	0.32 from the data sheet midpoint	audit F10
NSTAR I_{sp}, η	3150 s, 0.62	3270 s, 0.65	[4]
RTG decay τ	75 yr	40.1 yr (110→72 We in 17 yr)	[2]
RTG energy	finite times printed	"never" when $E > P_0\tau$	integral of Eq. 10
NEP mass	33 kg/kWe × 100 kWe lumped	AMTEC split model 3390 kg at 100.8 kWe	[8]
Radiator	27 m^2, T, ε unstated, 12 kg/m^2	T = 650 K, ε = 0.85, 7.67 kg/m^2	[8]
Solar array	200 kg (Table 1) / 450 kg (Figs. 5–6)	25 kg/kW	lecture notes (UNVERIFIED value)
Solar power, Table 2	undeclared 3.16 kW	10 kW at 1 AU, $1/r^2$ along the path	[12]
Mars cargo	α per kW thruster, no PPU/tanks/bus	AMTEC model + PPU + thrusters + 10 % tanks + 1 t bus	audit F4

4.2 Corrected analysis of the same missions (Case B)

Case B keeps the paper's mission definitions and applies only the audit corrections listed in Table 6; no redesign is done at this stage.

Results. With these corrections, the NEP 100 kWe craft of Table 1 needs 3.00 yr (Table 4). A solar craft of the paper's mass needs 33.8 yr at 1 AU, and at 5 AU or on any RTG the mission never completes. The ratio of solar to NEP time is therefore about 11, not 20. The advantage of NEP is real, but it comes from distance rather than from the nominal factor. Figure 2 shows that, at equal power-system mass, a solar array out-produces a fission system of the same mass out to the crossover distance marked in the figure for the 2020s system, and to about 1.1–1.4 AU for the 2030s and 2040s systems. An RTG system of the same mass is 3 to 15 times weaker than the fission system at every distance.

The Mars cargo case with the paper's definition. With every correction applied and the paper's inputs kept (5 t, 200 d, a 2000 s Hall thruster, $\eta = 0.5$ and a ΔV of 6.25 km/s from Eq. 27), the mission closes only at 250 kWe, 18.5 t initial mass and a 27 % payload fraction. The reference 100 kWe system cannot do it (closure probability 0 in 2000 Latin-hypercube samples; required power 236–369 kWe, 5th–95th percentile), and solar-electric propulsion does not close at all. Two further problems are independent of mass: 200 d is shorter than the 259-d Hohmann time, and the ΔV itself is not supported by any source. Section 4.3 uses the optimal-control solver to test whether 200 d is reachable at all.

4.3 Reconciling the two models

The simulation (`sim`) and the SMAD-style design study (`smad`) were built independently, and they disagreed in four places. (1) The simulation found that the 8 t, 450-d Mars cargo mission closes in every technology tier, while SMAD rated it not possible in the 2020s. (2) The simulation's trip times come from a tangential burn-coast-burn family, whereas SMAD uses a $\Delta V(T)$ law calibrated at one point. (3) Launch capacity versus C_3 and the margins were handled differently. (4) The 2030s and 2040s tiers of the simulation were parameterised placeholders. In this subsection we build one common input set, run both trajectory models on it together with the new optimal-control solver, and explain every remaining difference.

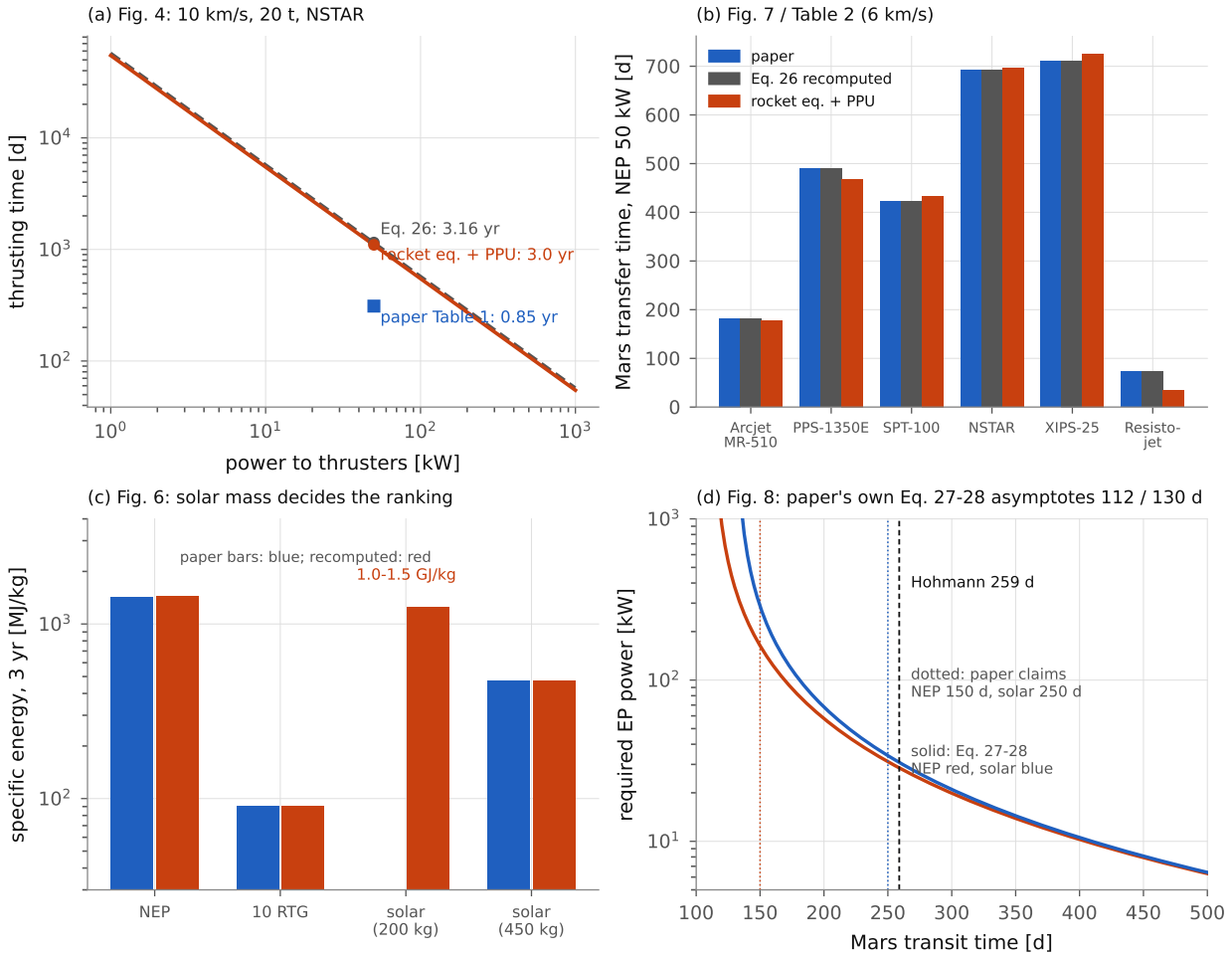


Figure 1: Corrected versions of the original Figs. 4, 6, 7 and 8. (a) NEP time versus thruster power: Eq. 26 and the rocket-equation form agree at the 5 % level for this case, but the printed Table 1 point (blue) is 3.7× too low. (b) NEP column of Table 2: the paper's values reproduce Eq. 26, but the resistojet and arcjet bars are not flyable. (c) Specific energy: the ranking of NEP and solar depends entirely on which solar mass is used. (d) Required electric power versus transit time from the paper's Eqs. 27–28: the asymptotes are at 112 and 130 d, not 150 and 250 d, and every point left of the dashed line is below the Hohmann flight time. None of the four original figures survives the corrections unchanged.

Common inputs. The missions are a Mars cargo of 8 t in 450 d ($C_3 = 8.2$ km^2/s^2), a Jupiter orbiter of 1.5 t in 6 yr ($C_3 = 40$) and a Uranus orbiter of 1 t in 10 yr ($C_3 = 40$). The planets are circular and coplanar, and arrival is defined as rendezvous with the planet's orbit followed by a capture spiral to 20 (Mars) or 30 (giants) planetary radii. The ΔV contingency is 5 %, the trip time is designed to 95 % of the limit, and the thrust duty is 0.88, 0.80 and 0.70. The thrusters, the PPU efficiency of 0.93, the bus load, the bus mass without the SMAD system margin, the 10 % reserve and the 10 % tanks are as in Section 2. The launch vehicles are SMAD's three-vehicle catalogue (identical in both packages: 13,011, 31,181 and 37,069 kg at $C_3 = 8.2$ for Falcon Heavy, SLS Block 1B and Starship) with the 1.15 reserve, and the power-system masses come from the tables of the three SMAD eras. The closure engine is the simulation's mass-closure function, and the three trajectory models are plugged into it through one interface.

The common closure reproduces SMAD. With the SMAD $\Delta V(T)$ law as the trajectory model, the common closure returns 57 kWe / 16.2 t for Mars in the 2030s against SMAD's own 60 kWe / 17.2 t, 17 / 5.7 against 18 / 6.1 for Jupiter in the 2030s, and 81 / 23.2 against 96 / 28.0 for Mars in the 2020s (Table 8). The method is therefore a faithful stand-in for the SMAD model to within 5–15 %, and any remaining difference below is due to the trajectory model or to the definitions, not to the closure code.

Finding 1: the simulation's Mars result for the technology tiers is an artefact of failed table nodes. The simulation tabulates burn-coast-burn outcomes against initial acceleration and exhaust velocity, and it fills the nodes where its solver failed with the nearest solved node. For Mars at $C_3 = 8.2$, 35 of the

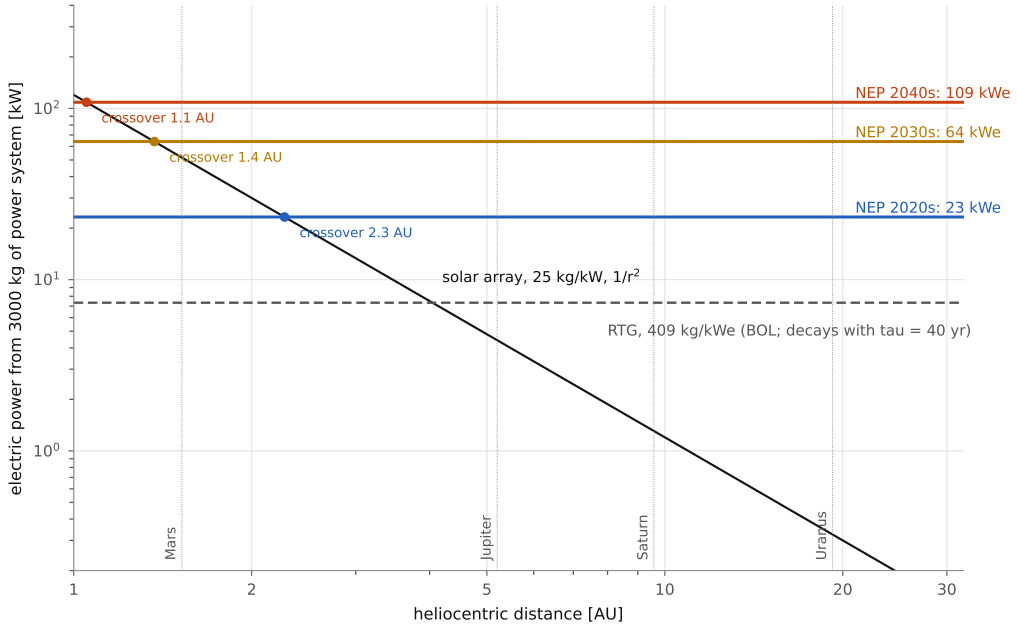


Figure 2: Electric power available from 3000 kg of power system versus heliocentric distance. Solar: 25 kg/kW, $1/r^2$. RTG: 409 kg/kWe at beginning of life. NEP: tier power-system tables of the SMAD study. Dots: crossover with the solar array. Beyond the crossover, the fission systems deliver the most power for a given mass.

63 nodes failed, and the whole range $a_0 < 6.3 \times 10^{-4}$ m/s^2 is filled. All NEP designs of interest ($a_0 \approx 10^{-4}$) lie inside this range, where the table returns the impulsive Hohmann values (about 300 d, propellant fraction 0.07–0.08) independently of thrust. The simulation's Mars rows for all tiers (trip 396–402 d, "closes in every tier", P(close) 0.55–0.68) are therefore not results of its trajectory model (errata E-S0). When these nodes are treated as "no solution", the burn-coast-burn family has no Mars solution at $C_3 = 8.2$ for these accelerations, because the family cannot represent a low-thrust transfer that starts with excess speed. The optimal-control solver has no such failure. It finds a 428-d trajectory with 64 kWe in the 2030s and 95 kWe in the 2020s, in agreement with SMAD (60, 96 kWe).

Finding 2: burn-coast-burn is not far from optimal in propellant but is slow. Table 7 and Fig. 3 compare the two families for the same vehicle. At equal trip time, the optimal propellant fraction equals the burn-coast-burn value to within -0.9–0.7 %, because both reach the impulsive plateau. The minimum time of the optimal solution, however, is 3–46 % shorter (median 30 %), and the gap grows with initial acceleration: a burn-coast-burn transfer cannot go below roughly the Hohmann time, whereas continuous steering can. The family is therefore a safe upper bound for the trip time with no more than a few percent propellant penalty, and the simulation's description of it as an upper bound is correct. Figure 5 shows the same effect at mass closure, as trip time against electric power.

Finding 3: the paper's 200-day Mars mission is not reachable. With the paper's thruster (Hall, 2000 s), $C_3 = 0$ and optimal steering, a heliocentric time of 200 d needs an initial acceleration of 6.3e-04 m/s^2 and a propellant fraction above one half ($\Delta V \approx 15$ km/s, not the 6.25 km/s of Eq. 27). With the paper's mass convention of 33 kg per kW of thruster power, the power-system mass alone at that acceleration is $0.41\, m_0$, and the mass equation has no positive solution at any power (none). The simulation's statement that this mission "closes at 250 kWe" (Case B) follows from the unsupported Eq. 27 and is withdrawn.

Finding 4: the Jupiter and Uranus disagreement is a definition. SMAD's reference ΔV for Jupiter (10 km/s at $C_3 = 30$) contains about 1 km/s of capture, whereas the common definition (rendezvous plus a spiral to 30 planetary radii) needs 7.7 km/s for the spiral alone. With the common definition, the Jupiter 2030s design needs 45 kWe and 9.4 t, against SMAD's 18 kWe and 6.1 t; with a 1 km/s capture, the same closure gives 29 kWe (Table 16). The burn-coast-burn family gives 52 kWe / 9.6 t, within about 15 % of the optimal-control value. This difference favours SMAD's verdicts for the outer planets and is the largest unresolved modelling choice of the study (errata E-M2).

Table 7: Burn-coast-burn (sim) versus optimal control for a constant-power vehicle, I_{sp} 4170 s. "solver failed" = the sim table node was filled with a neighbour's value (errata E-S0).

Target	C_3	a_0 [m/s^2]	T_{bcb} [d]	$T_{opt,min}$ [d]	time saved [%]	ϕ_{bcb}	ϕ_{opt} at T_{bcb}
Mars	0	5e-05	1321	1288	3	0.129	0.130
Mars	0	1e-04	782	711	9	0.129	0.129
Mars	0	2e-04	420	381	9	0.128	0.128
Mars	0	4e-04	338	270	20	0.128	0.128
Mars	0	8e-04	297	194	34	0.128	0.128
Mars	8.2	5e-05	solver failed	962	-	-	-
Mars	8.2	1e-04	solver failed	434	-	-	-
Mars	8.2	2e-04	solver failed	292	-	-	-
Mars	8.2	4e-04	solver failed	224	-	-	-
Mars	8.2	8e-04	277	171	38	0.065	0.065
Jupiter	0	1e-04	2506	2073	17	0.332	0.333
Jupiter	0	2e-04	1759	1216	31	0.328	0.328
Jupiter	0	4e-04	1265	715	43	0.315	0.315
Jupiter	40	5e-05	solver failed	2329	-	-	-
Jupiter	40	1e-04	1556	1162	25	0.200	0.199
Jupiter	40	2e-04	1180	820	30	0.188	0.188
Jupiter	40	4e-04	1083	587	46	0.182	0.184

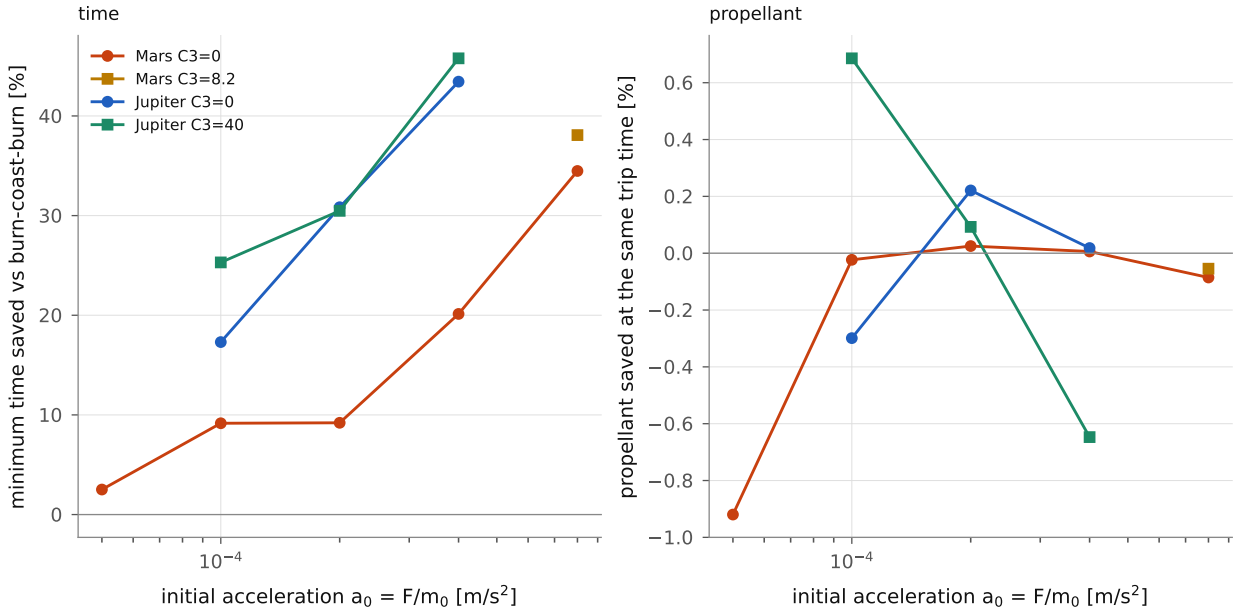


Figure 3: Optimal control versus the burn-coast-burn family: percentage of minimum time (left) and of propellant at equal trip time (right) saved, against initial acceleration, for Mars and Jupiter at two launch energies (only table nodes where the simulation solver succeeded). The saving is almost entirely in time, not in propellant.

Finding 5: bookkeeping differences in the simulation. The simulation's technology-tier bus mass includes SMAD's 10 % system margin and then applies a second 10 % reserve, while it omits the integration allowance through a stale ledger key; as a result, the bus is 4329 kg against 3278 kg for the 2020s Mars design (errata E-S1, E-S2). The closure-probability conventions also differ (fixed hardware versus re-optimised in each sample), which is why the same Mars design had 0.55 in one study and 0.27 in the other.

Which model is more defensible? For trajectories, the optimal-control model is the more defensible: it gives a feasible solution with no table failures, and it bounds both other models. For the mass model, SMAD's bottom-up ledger (maturity-based margins, structure and thermal floors) is more complete than the simulation's frozen bus plus reserve, but its shield and conversion models are conceptual (Section 3). The two models cannot be reconciled on the outer-planet capture definition, so both numbers are reported. Figure 4 summarises the power and mass of the minimum-mass designs from all models.

4.4 Feasible redesign: what changes and why

Having shown that the original mission cannot be closed as defined, we now look for the lightest design that does close, first with the simulation alone and then for all three missions on the common inputs.

Table 8: Reconciliation: P_e [kWe] / escape or launch mass [t] / trip time [d] of the minimum-mass design in each model. "none" = no closure within the power grid (8–600 kWe); "n/a" = no table.

Mission	Era	SMAD native	sim native	common + dV(T) law	common + burn-coast-burn	common + optimal control
Mars cargo 8 t, 450 d	2020s	96 / 28.0 / 427	63 / 21.4 / 402	81 / 23.2 / 427	none	95 / 24.9 / 428
	2030s	60 / 17.2 / 427	49 / 15.3 / 396	57 / 16.2 / 428	none	64 / 16.9 / 428
	2040s	62 / 15.0 / 427	43 / 13.8 / 400	60 / 14.4 / 427	none	80 / 14.9 / 428
Jupiter orbiter 1.5 t, 6 yr	2020s	23 / 9.7 / 2082	72 / 17.7 / 1866	22 / 8.7 / 2086	86 / 19.3 / 2081	89 / 20.4 / 2082
	2030s	18 / 6.1 / 2082	38 / 8.9 / 1882	17 / 5.7 / 2084	52 / 9.6 / 1809	45 / 9.4 / 2082
	2040s	15 / 5.2 / 2082	33 / 7.4 / 1819	16 / 5.1 / 2082	52 / 8.1 / 1627	40 / 7.6 / 2082
Uranus orbiter 1 t, 10 yr	2020s	26 / 11.7 / 3469	none	23 / 9.7 / 3475	none	none
	2030s	17 / 6.2 / 3469	none	16 / 5.8 / 3472	none	105 / 17.9 / 3469
	2040s	15 / 5.4 / 3469	none	15 / 5.1 / 3465	none	85 / 11.2 / 3469

Table 9: Values Changed from the Corrected Paper Case (B) to the Redesign (C) (source: `sim/changes_log.md`).

Item	Old (B)	New (C)	Reason
Trip time	200 d (dV 6.25 km/s, below the 259-d Hohmann time)	492 d, full integration incl. capture	the 200-d definition has no source
Thruster	Hall 2000 s, η 0.50	NEXT 4170 s, η 0.706, 9 units	life 51,184 h per unit [6]; count sized for $\leq$80 % life
Power	100 kWe reference	71.9 kWe	minimum initial mass under the policy
f_{EP}	0.5	0.93 ($= 1 - 5\,\mathrm{kWe}/P_e$)	half the power on the bus is not credible at tens of kWe
Reserve	none	15 % on bus, power, thrusters	early-design growth (assumption)
Isp	2000 s	4170 s; optimum of the generic fit 3700 s	power-limited trade, flat to 0.1 % over 3600–4170 s
Escape mass	9.5 t	11.8 t at $C_3 = 0$	output of the closure

Case C (simulation, Earth escape at $C_3 = 0$, 5 t payload). Starting from the corrected Case B, a deterministic grid search was run over thruster (NEXT, NSTAR, HERMeS) and electric power (40–2000 kWe) for the minimum initial mass under a margin policy: initial mass $\leq$90 % of the advisory 20 t limit, trip $\leq$85 % of 600 d, thruster life and throughput $\leq$80 % of demonstrated, and a radiator within the largest studied. The search gives a NEXT cluster of 71.9 kWe with 11,797 kg at escape, 1715 kg of xenon and 2748 kg of power system (radiator 19 m^2, 242 kW rejected). It uses nine thrusters at 0.18 of the demonstrated life and flies a 492-d trip (burn 228 d, coast 101 d, burn 118 d, capture 45 d; ΔV 5.63 plus 0.79 km/s). Table 9 lists the values changed and the reason for each. In a Monte Carlo with the hardware fixed, the closure probability is 0.74, and the failure mode is propellant mass, driven by the specific-mass range of 28.7–45.6 kg/kWe. This case uses the $C_3 = 0$ trajectory table, which has no failed nodes, so it is unaffected by the defect described in Section 4.3.

Era designs on the common inputs. Table 13 gives the mass ledger of the minimum-mass designs of the three missions in the three eras, computed with the common inputs and the optimal-control trajectory model (Section 4.3). Figure 6 shows why the era matters. The payload fraction at the optimum I_{sp} falls steeply with power-system specific mass, so in the Mars case the 2020s power system (about 80 kg/kWe at 100 kWe) leaves 0.6 of the launch mass for payload and bus, the 2030s system (42 kg/kWe) leaves 0.72 and the 2040s system (27 kg/kWe) leaves 0.78. For Jupiter, the corresponding fractions are 0.41, 0.56 and 0.63. Figure 7 shows how the optimum I_{sp} itself moves with the era.

4.5 Thermal and radiator design

Every kilowatt of electric power leaves a fission system with four to ten kilowatts of heat that can only be radiated, so the radiator, not the reactor, sets the geometry of the vehicle. The SMAD study sizes it with the emitted flux $q = \varepsilon\sigma F\eta_{fin}(T_r^4 - T_s^4)$ per emitting surface and $A = 1.10\,Q_{rej}/q$, where the 10 % margin covers heat-pipe and meteoroid losses and the area counts both faces of two-sided panels, as in [11, 13]. The study uses $\varepsilon = 0.85$, $F = 0.95$ and $\eta_{fin} = 0.90$, with a sink temperature for a flat panel within 30° of the Sun line of $T_s^4 = \alpha S(r)\sin\beta/(2\varepsilon\sigma) + T_{cmb}^4$ ($\alpha = 0.25$ at end of life). The sink temperature is 205 K at 1 AU, 90 K at Jupiter, 47 K at Uranus and 37 K at Neptune. As a result, the design is set by the hottest case (1 AU, early in the mission), and the outer planets only add margin in the cold case.

Temperature trade. Raising the radiator temperature shrinks the area as $1/(T_r^4 - T_s^4)$, but it also lowers the cycle efficiency $\eta_{cyc} = f_C(1 - T_r/T_{hot})$, which inflates the reactor, shield and fuel. Figure 8 shows the

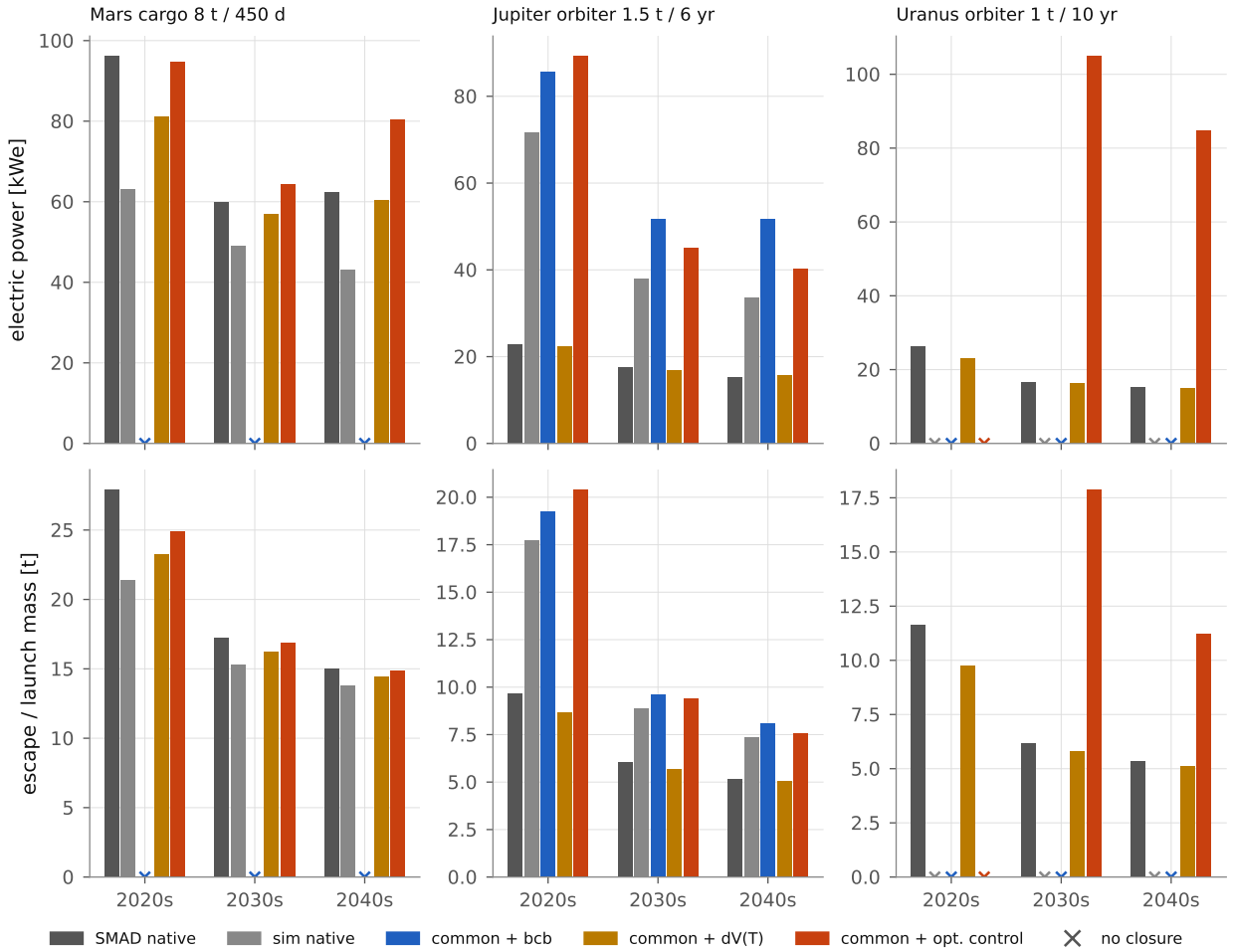


Figure 4: Power (top) and mass (bottom) of the minimum-mass designs in the SMAD study, the simulation, and the common closure with three trajectory models. Crosses: no closure (the burn-coast-burn family has none for Mars; the simulation's own Mars bars are artefacts, E-S0). For Mars the common closure and SMAD agree; for Jupiter they differ through the capture definition.

result for the 17.5 kWe Jupiter orbiter. There is a shallow interior minimum of power-system mass at about 550 K for Brayton, Stirling and AMTEC conversion (1346, 1315 and 1274 kg) and at about 790 K for silicon-germanium thermoelectrics (1783 kg), with radiator areas of 27, 19, 14 and 76 m^2 at the respective minima. For an ideal Carnot machine with a fixed-size radiator the optimum would be $0.75\,T_{hot}$ (862 K for 1150 K). The lower real optimum reflects the Carnot fraction $f_C < 1$ and the step in radiator areal density from 3.8 kg/m^2 (composite panels with water heat pipes, 375–550 K, COMPASS [11]) to 7.67 kg/m^2 for armoured refractory heat pipes above 650 K [8]. A Brayton cycle optimised for 540 K, as in [13], is an input assumption, and holding the Carnot fraction constant as T_r rises is optimistic.

Radiator numbers per case. The advanced-tier designs radiate 307 kWt (Mars cargo, 98 m^2 at 540 K, 504 kg), 90 kWt (Jupiter, 29 m^2, 147 kg) and 86 kWt (Uranus, 27 m^2, 140 kg). The near-term Mars stack needs 956 kWt at 720 K (96 m^2, 1342 kg) because its power-system efficiency is lower. All of these are far below the 2,500 m^2 of the largest studied system [11]. Since the area does not exceed 10^2 m^2 in any case, the issue is deployment rather than area: panels of 4×5 m form two wings of 24–34 m inside the shield umbra. The areal densities used are anchored to 3.8 kg/m^2 (COMPASS), 8.8 kg/m^2 (JIMO Brayton, [13]) and 7.67 kg/m^2 [8], with a 0.55–1.6 multiplier covering the spread down to the 2.2 kg/m^2 carbon-carbon goal [15], which is a development goal and not a measurement.

Bus thermal control. The PPU waste heat (4.0 kW for Mars, 0.8 kW for the outer planets) is rejected on a 330 K cold-plate radiator (26 kg at Mars). MLI at 0.73 kg/m^2 and 2–5 % of the dry mass for the thermal subsystem follow SMAD [5]. Radiation from the hot wings onto the bus is only about 1 W, because the wings are edge-on; a 10° tilt raises it by 10–30×, which is still only tens of watts. In the outer solar system, a cold bus is cheap to keep warm with a fission source, since the kilowatt-class electric budget dwarfs the

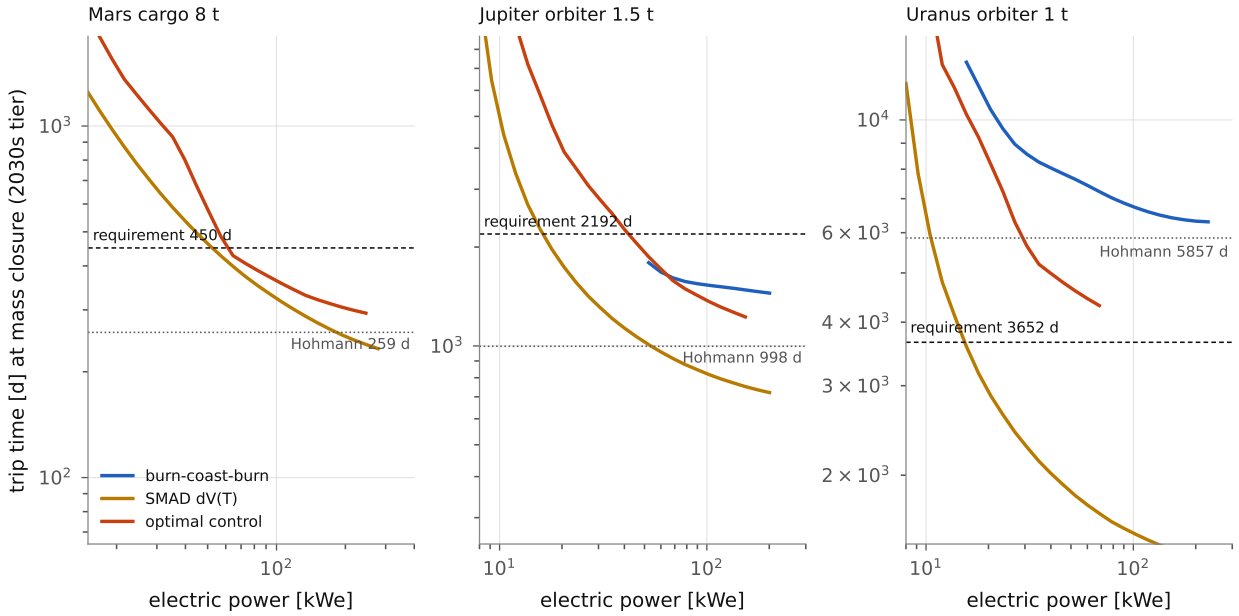


Figure 5: Trip time versus electric power at mass closure (2030s tier, thrust-always-on fastest solution for the optimal-control curve), with the requirement and the Hohmann flight time. Where the burn-coast-burn curve is absent, the family has no solution.

Table 10: Mass ledger [kg] of the minimum-mass optimal-control designs (common inputs); radiator in m^2, thruster life as fraction of the demonstrated NEXT value, times in days.

Mission	Era	P_e	m_{ps}	m_{thr}	m_{bus}	reserve	tanks	Xe	m_0	A_{rad}	N_{thr}	life	T_{helio} / T_{cap}
Mars cargo 8 t, 450 d	2020s	95	7685	747	3278	1171	366	3655	24901	94	13	0.21	363 / 65
	2030s	64	2935	511	2147	559	248	2475	16875	106	9	0.21	363 / 65
	2040s	80	2276	633	1966	487	140	1396	14898	44	11	0.14	370 / 57
Jupiter orbiter 1.5 t, 6 yr	2020s	89	7594	681	1564	984	737	7370	20429	191	12	0.78	1474 / 609
	2030s	45	2321	327	1160	381	338	3381	9408	74	6	0.77	1478 / 605
	2040s	40	1483	280	1133	290	262	2617	7564	29	5	0.68	1517 / 566
	2030s	105	4544	855	1162	656	880	8804	17902	173	16	0.79	3331 / 138
	2040s	85	2500	638	1138	428	501	5010	11214	61	11	0.64	3351 / 119

tens-of-watt MLI leak; this is an advantage over RTG-powered probes. This part of the analysis is at the differential-element level, and a full thermal model is still needed.

4.6 SMAD subsystem sizing and technology readiness

The SMAD-style study sizes each vehicle bottom-up, following Larson and Wertz [5]: requirements lead to a functional breakdown, then to sizing equations, and finally to mass and power budgets with maturity-based margins. The 2011 edition was not available, and nothing is taken from it. The margins are a 25 % mass-growth allowance on new equipment (reactor, converter, PMAD, radiator, shield, boom, harness), 15 % on the PPU, feed, structure, ADCS, communications, C&DH, thermal and auxiliary power (an interpolation inside the SMAD range), 5 % on flown thrusters and tanks, 2 % for integration and a 10 % system margin. The usual pre-Phase-A system margin in concurrent engineering is 30 % [16]; the lower value is kept because the item margins are already large. Power loads carry 20 %, ΔV carries 5 %, and a 15 % launch reserve is applied. Structure is at least 8 % of the injected mass and thermal control at least 2 % of the dry mass (SMAD Table 10-10).

Subsystems. The *reactor* mass is $m = 436 + 0.1195\,P_{th}$ kg (fitted to AMTEC-SNRP and SP-100) with a 25 % margin; the burnup is 0.2–0.4 % against a 3 % limit, so the fuel is not a constraint. The baseline *conversion* is a closed Brayton cycle (He-Xe, 1150 K inlet, 21.7 % cycle, 7.2 kg/kWe converter, TRL 5, about 50,000 h of ground test), with Stirling, AMTEC and SiGe thermoelectrics as alternatives; the last is the only flight-proven conversion, but it costs 13–17 % more launch mass. The *shield* is a LiH/W cone sized to a dose limit of 25 krad(Si) and 10^{11} n/cm^2 at the payload over the mission. It comes from a point-source exponential-removal model that reproduces the JIMO shield to within a factor of 1.9 (conservative), and the shield half-angle and boom length are optimised jointly (2–3°, 31–41 m). For *propulsion*, NEXT-class ion thrusters are used (6.9 kW, 13.5 kg assumed) with a spare count of max(2, 15 %), a PPU of 5 kg/kW sized

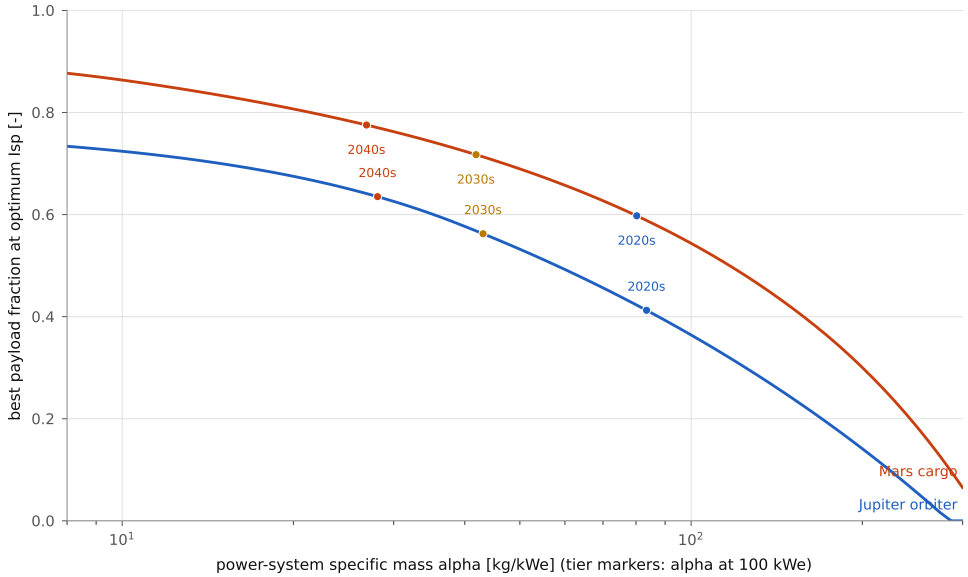


Figure 6: Best payload-plus-bus fraction (at the optimum specific impulse) versus power-system specific mass. Markers: the specific mass at 100 kWe in each era. A lighter power system leaves a markedly larger fraction for payload and bus.

Table 11: Readiness of the Critical Items in 2026 (SMAD study; the TRL values are our judgement from the cited histories).

Item	TRL	Demonstrated vs missing
Fission reactor, 100–500 kWt, years at full power	4	KRUSTY 1 kWe ground test 2018 [17]; SP-100 ground tests, never flown [8]. Missing: a flight-qualified reactor with multi-year fuel qualification.
Brayton conversion	4–5	10 kWe Brayton unit about 50,000 h on the ground; none has run in space [13, 14]. Missing: 100 kWe unit and a 100,000-h test.
Heat-rejection, 30–100 m^2 at 540 K	4	Sub-scale and full-scale panels, NaK loops to 875 K [14]. Missing: deployment of 100s of m^2 in vacuum.
Shield (LiH/W)	4	Ground-test history; LiH swelling and hydrogen containment problems found under Prometheus. Missing: flight-like thermal management.
PMAD ∼1 kV and direct drive	4–5	960 Vac concept [14]; direct-drive Hall units TRL 3–4 [18, 19].
Thrusters (NEXT-class ion)	6	51,184 h and 918 kg Xe [6]; HERMeS 3,570 h wear test [20].
Boom, ADCS, nuclear launch approval	5	Analysed, not flown; US launched only SNAP-10A.

1.15× for a spare string, and xenon tanks at about 10 % of the propellant (carbon-overwrapped; krypton and argon are heavier and lose efficiency). The *structure, ADCS, communications and C&DH* comprise a 30–45 m boom with a first bending mode ≥0.1 Hz, thrust-offset torque unloaded by thruster gimbals, a 4 m Ka-band antenna and rad-tolerant processors.

Technology readiness. Table 11 summarises the readiness of the critical items today. The reactor, converter, radiator, shield, PMAD and autonomous operation all sit at TRL 4–5, and only the thrusters, xenon storage and C&DH are at TRL 6 or above. The binding quantity is the converter life: a full-power life of 3, 8 and 12 yr is 0.5, 1.4 and 2.1× the 50,000-h demonstration.

Mass budgets. Figure 9 shows the SMAD launch-mass breakdown by case and era. The power system is 20–45 % of the dry mass and shrinks from 8 t (2020s Mars) to 2 t (2040s), the propellant is 13–33 % of the launch mass, and the margins plus integration are 8–10 %.

Development plan by tier. The 2030s state needs a flight-like reactor and shield ground test, a full-scale deployed radiator, a 100-kWe-class converter and, as the critical path, a 100,000-h (11.4-yr) real-time converter life test starting in 2029. The flight unit is then a 2044 article, or a 2038 article if 50,000 h is accepted, at the cost of a 1.4–2.1× gap for the outer-planet missions. The 2040s state adds a 1500-K Brayton with a new fuel form and refractory alloys, a 2.2 kg/m^2 carbon-fibre radiator, 100 krad / 10^{13} n/cm^2

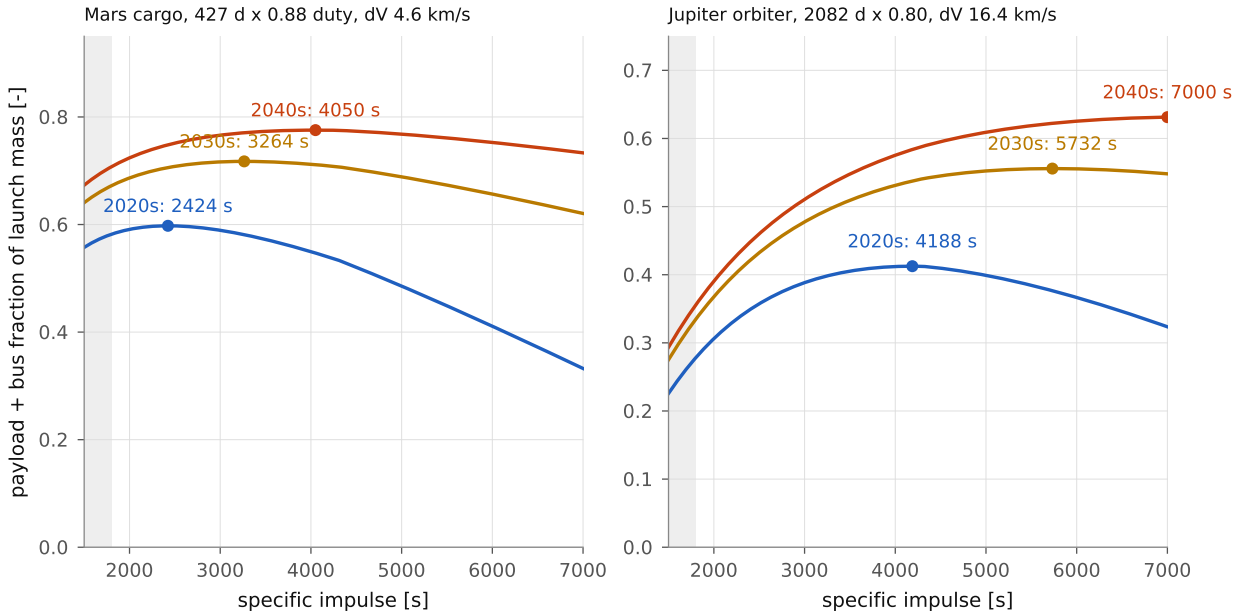


Figure 7: Payload-plus-bus fraction versus specific impulse in three eras for the Mars and Jupiter cases (power-limited model with the fitted thruster efficiency; above 4300 s the efficiency is held at its 4300 s value, so the 2040s Jupiter optimum at 7000 s is an extrapolation). Dots mark the optimum.

Table 12: Final verdict table: common-input closure with the optimal-control trajectory model. P_e and m_0 are the minimum-mass design at nominal inputs; P(close) is the Monte Carlo probability (N = 2000, seed 20251103) that *some* power closes the mission when the design is re-optimised in every sample; the percentile column is the sampled minimum-mass power. TRL = lowest critical TRL of the era; life ratio = required full-power converter life over the demonstrated hours.

Mission	Era	P_e [kWe] / m_0 [t]	Trip [d]	P(close)	P_e 50 % (5–95) [kWe]	TRL / life ratio	Verdict (this paper)	SMAD verdict
Mars cargo 8 t, 450 d	2020s	95 / 24.9	428	1.00	103 (91–133)	4 / 0.5	Marginal	Not possible
	2030s	64 / 16.9	428	1.00	70 (62–80)	6 / 0.3	Possible	Possible
	2040s	80 / 14.9	428	1.00	80 (62–91)	5 / 0.2	Marginal	Marginal
Jupiter orbiter 1.5 t, 6 yr	2020s	89 / 20.4	2082	0.51	103 (80–117)	4 / 1.4	Marginal	Marginal
	2030s	45 / 9.4	2082	1.00	54 (48–62)	6 / 0.7	Possible	Possible
	2040s	40 / 7.6	2082	1.00	48 (37–54)	5 / 0.5	Marginal	Marginal
Uranus orbiter 1 t, 10 yr	2020s	none	-	0	-	4 / 2.1	Not possible	Not possible
	2030s	105 / 17.9	3469	0.63	133 (103–151)	6 / 1.1	Marginal	Marginal
	2040s	85 / 11.2	3469	0.99	103 (80–151)	5 / 0.7	Marginal	Marginal

avionics qualification, a 30,000-h high-I_{sp} thruster test and an operational refuelled Starship. Its critical path is again the converter (150,000 h of partly accelerated evidence), and the flight unit is a 2050 article. Figure 10 gives both schedules; the durations are assumptions except where a test length is fixed by physics. The tier names (2020s, 2030s, 2040s) are technology *states*, not forecasts: they state what must be true, and when it could be true if funded from 2027.

4.7 Mission results by era

Table 12 is the final verdict table. It combines the physical closure, computed with the common inputs and the optimal-control trajectory model, with the readiness of the era through the SMAD rule of Section 2. The closure is given as the power and mass of the minimum-mass design and the probability that *some* power closes the mission when the design is re-optimised in every Monte Carlo sample; the readiness is given as the lowest critical TRL and the required converter life over the demonstrated hours. Figure 11 shows the same matrix, and Fig. 12 shows the trajectories of the 2030s Mars and Jupiter designs.

Mars cargo, 8 t in 450 d. The mission closes in every era: 95 kWe and 24.9 t in the 2020s, 64 kWe and 16.9 t in the 2030s, and 80 kWe and 14.9 t in the 2040s (trip 428 d against the 450-d limit; Hohmann flight time 259 d). The three trajectory models agree here, and SMAD's native 2030s design is 60 kWe / 17.2 t. The first Mars verdict that changes is the 2020s one, where the simulation reported "closes" and SMAD "not possible". On common inputs the *physics* closes (95 kWe and 24.9 t at escape, within the 31 t of one SLS Block 1B at $C_3 = 8.2$), but the readiness of the era (critical TRL 4) keeps it Marginal rather than possible. The 100 kWe system of the original paper would be needed only for much faster trips, and

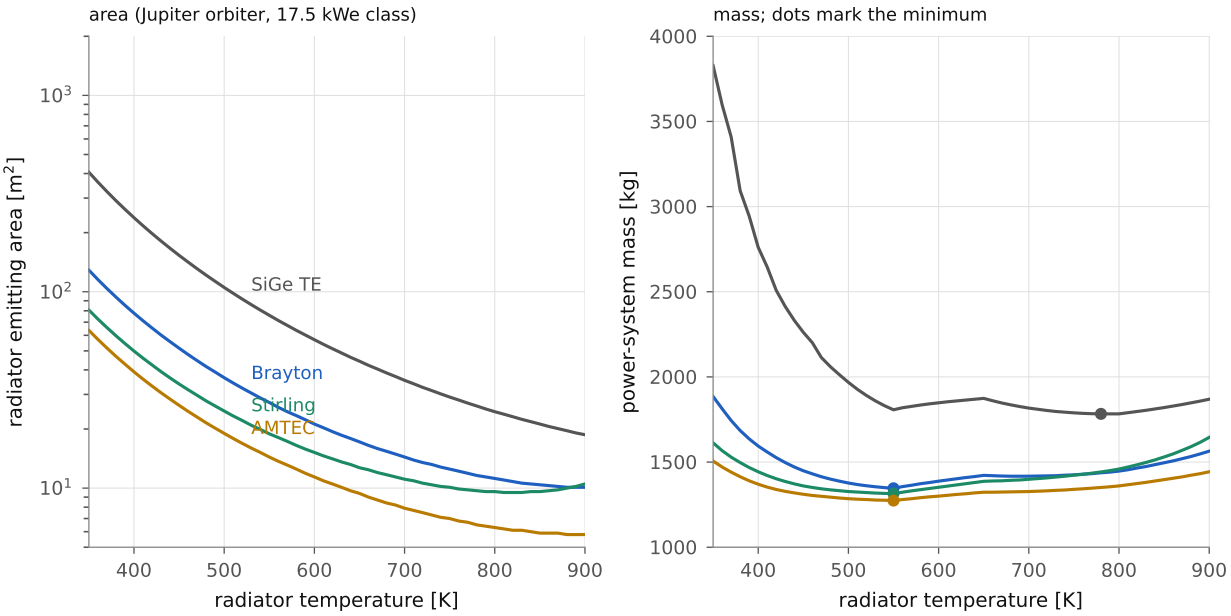


Figure 8: Radiator emitting area (left) and total power-system mass (right) versus radiator temperature for four conversion technologies at the Jupiter-orbiter design power (SMAD study). Dots: mass minimum. The mass minima are shallow for all four technologies.

Table 13: Mass ledger [kg] of the minimum-mass optimal-control designs (common inputs); radiator in m^2, thruster life as fraction of the demonstrated NEXT value, times in days.

Mission	Era	P_e	m_{ps}	m_{thr}	m_{bus}	reserve	tanks	Xe	m_0	A_{rad}	N_{thr}	life	T_{helio} / T_{cap}
Mars cargo 8 t, 450 d	2020s	95	7685	747	3278	1171	366	3655	24901	94	13	0.21	363 / 65
	2030s	64	2935	511	2147	559	248	2475	16875	106	9	0.21	363 / 65
	2040s	80	2276	633	1966	487	140	1396	14898	44	11	0.14	370 / 57
Jupiter orbiter 1.5 t, 6 yr	2020s	89	7594	681	1564	984	737	7370	20429	191	12	0.78	1474 / 609
	2030s	45	2321	327	1160	381	338	3381	9408	74	6	0.77	1478 / 605
	2040s	40	1483	280	1133	290	262	2617	7564	29	5	0.68	1517 / 566
	2030s	105	4544	855	1162	656	880	8804	17902	173	16	0.79	3331 / 138
	2040s	85	2500	638	1138	428	501	5010	11214	61	11	0.64	3351 / 119

the 200-d, 5 t Mars cargo of the original is not closable (Section 4.3).

Jupiter orbiter, 1.5 t in 6 yr. Under the common definition (rendezvous with Jupiter's orbit and a spiral to 30 R_J), the 2030s design is 45 kWe and 9.4 t (Possible; P(close) 1.00), the 2020s design 89 kWe and 20.4 t (Marginal) and the 2040s design 40 kWe and 7.6 t (Marginal). SMAD's native results are 3–4× lighter in power (18 kWe and 6.1 t in the 2030s), because its reference ΔV counts about 1 km/s of capture (errata E-M2); with the same capture assumption, the common closure gives 29 kWe. Figure 12 shows the 2030s trajectory: the vehicle thrusts to day 558, coasts for 512 d, thrusts again to day 1398 and makes a final 80-d coast to Jupiter's orbit at day 1478, followed by a 604-d capture spiral. The optimiser therefore coasts for a third of the heliocentric leg.

Uranus orbiter, 1 t in 10 yr. In the common planar model the transfer is direct, and the 10-yr limit is much shorter than the 16-yr Hohmann time. The closure therefore needs 105 kWe and 17.9 t in the 2030s (Marginal; P(close) 0.63) and 85 kWe in the 2040s, and the 2020s system does not close at any power in the grid. SMAD's 17 kWe rests on a literature trajectory with gravity assists (14.2 km/s of electric ΔV) that a planar direct-transfer model cannot represent; the fastest published Uranus times with 25 kWe-class NEP are about 11 yr for a 300 kg payload [21]. The Uranus rows of Table 12 are therefore a *conservative direct-transfer bound*, not a mission prediction, and the optimal-control table for Uranus is the least converged of the three (part of its a_0 range is extrapolated).

Solar versus fission, equal mass. To compare the power sources directly, the same 8 t Mars mission is flown in the animation data set `power_source_compare` with fission NEP, with solar-electric power (25 kg/kW, $1/r^2$) and with an RTG power system of the same mass (about 3 t). The solar craft reaches Mars orbit in 393 d against 362 d for the fission craft, and the RTG craft (7 kWe) does not get there. At

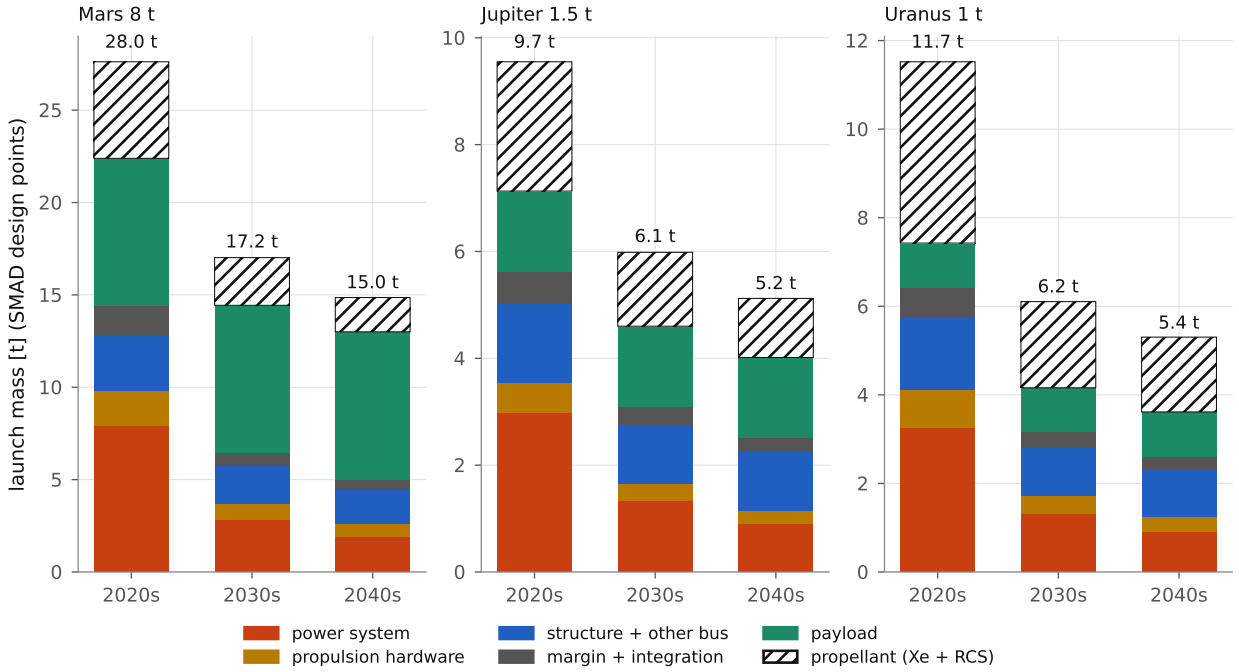


Figure 9: Launch-mass breakdown of the three SMAD design cases in the three technology eras (SMAD native design points, predicted masses with margins). Labels give total launch mass.

Mars, solar-electric propulsion is therefore nearly competitive; the advantage of fission is distance (Fig. 2), which is the same conclusion as that of the audit.

Where the models differ from the SMAD verdicts. Eight of the nine verdicts of Table 12 equal SMAD's (last column), although the Jupiter and Uranus powers differ by the magnitudes discussed above. The ninth is the 2020s Mars cargo, which is Marginal here and "not possible" in SMAD, because its Monte Carlo found only 27 % closure with a 28 t stack on an assumed Starship curve. The Mars 2040s power is also higher here than in SMAD (80 against 62 kWe at nearly equal mass): the minimum-mass optimum is flat in power, and the common optimal-control closure picks the faster end. The Mars difference is a difference of convention, since the common Monte Carlo lets the design power be re-chosen in each sample and uses the largest of three catalogue vehicles; a user who insists on SLS only or on Falcon Heavy only changes it (Table 17).

4.8 Novel methods: what holds

Design ideas were collected from three places: the simulation's novel-methods case (case E), the option screen of the SMAD study (its section 7a, a paired Monte Carlo of 17 uncertain inputs with $N = 177$ and the keep rule of a median gain $> 2\,\%$ and $P(\text{lighter}) \geq 0.8$), and the reconcile runs of this paper. Each idea was then checked against the literature with a strict rule: a source counts only if its text or abstract page was opened and the statement seen (Appendix A).

Kept. Table 14 lists the options that pass the screen. Three points limit how far they can be trusted. (i) Every gain is a *mass* gain on a power system that is only 20–45 % of the dry mass; the gains are not additive, and none removes a technology gate of Section 4.6. (ii) Two of the largest (the liquid-droplet radiator and the 1500 K Brayton) are TRL 3 research options with no flight or long-duration data [14, 22]. The options that are both mass-positive and at TRL ≥ 4 are the carbon-fibre radiator, radiation-hardened avionics, the xenon-tank gamma credit, PPU-heat reuse and a battery for peak loads. (iii) Most are not new. The literature check found precedent for the xenon-tank shield (mercury propellant in a NEP tug shield [23], and xenon and tanks credited in a Mars NEP shield [14]), the droplet radiator [24], the high-temperature Brayton [14] and the carbon-carbon radiator goal [15]. What this paper adds is the quantification on one consistent mission set, not the ideas themselves.

Simulation case E (quantified against the Mars and Jupiter designs).

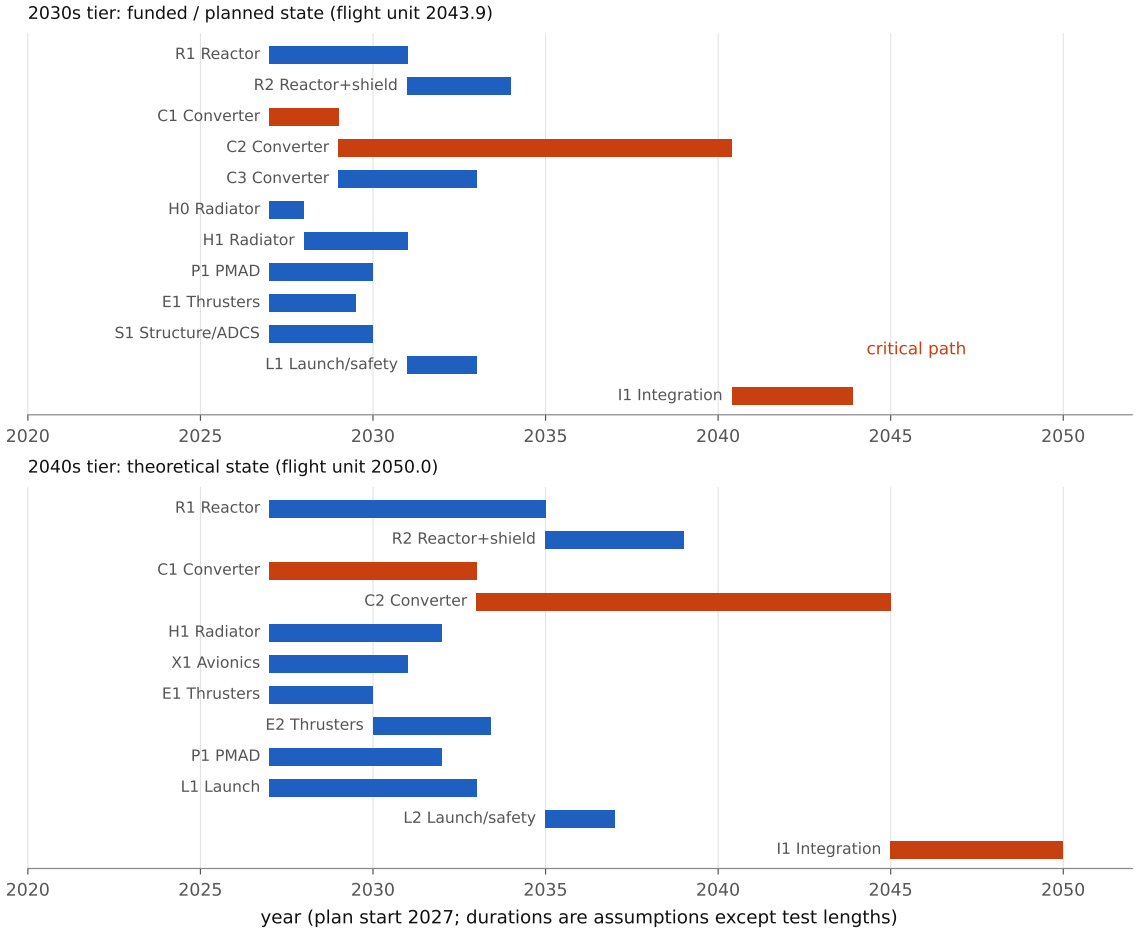


Figure 10: Development schedules implied by each tier (infinite-resource earliest finish, plan start 2027). Red: critical path, which is the converter life test in both tiers.

- *Free specific-impulse optimum* (precedent: the Stuhlinger velocity and optimum-I_{sp} results in [3], and analytic optima for ion, MPD and arcjet vehicles in an abstract-only source [25]). For the Mars cargo case at a 364-d trip, the free-I_{sp} optimum (2400 s, 141 kWe) delivers 5800 kg against 5668 kg for HERMeS (2802 s) and 5537 kg for NEXT (4170 s), a gain of +2.3 % and +4.7 % respectively. At 430–500 d the optimum moves to 3600 s, and the gain over NEXT falls to 1.5–2.4 %. Figure 7 shows that, as the power system gets lighter, the Mars optimum moves from about 2400 s (2020s) to 4050 s (2040s), and for the 2040s Jupiter case it moves beyond the 4300 s limit of the efficiency fit, where the number is an extrapolation. **Verdict: a known rule, kept as a design rule; the gain is small.**
- *Jettisoned propellant tanks* (staged tanks; [26], abstract only, reports about 10–13 % lower required specific mass for a power-limited vehicle at a payload ratio of 0.1). Halving the tank mass fraction lowers the initial mass by 0.9 % for Mars and 4.2 % for Jupiter (paired band −4.5 to −4.0 %). **Verdict: supported for the long, propellant-heavy Jupiter case and negligible for Mars; it costs a separation mechanism, and precedent exists.**
- *Jupiter gravity assist for Saturn and Uranus* (patched-conic upper bound). The electric ΔV falls from 15.7 to 10.3 km/s for Saturn (5.4 km/s saved) and by 4.8 km/s for Uranus, while the flight time roughly doubles (2208 to 4651 d for Saturn); the phasing (Jupiter–Saturn synodic period 19.9 yr) is not modelled. The literature check found no published electric-ΔV saving of this kind for NEP. It found chemical Jupiter-assist routes to Uranus of about 13 yr that are available only in certain launch windows [27, 28], and published NEP trajectories that use an Earth assist instead [21]. **Verdict: reported as a bound only, not claimed.**
- *Multiple reactors.* Using 2, 3 and 4 units raises the initial mass by 9.7, 16.7 and 22.4 % (Mars) for single-fault tolerance, and losing one of two reactors doubles the Mars trip (555 to 1070 d). Precedent: [29] (about 2–4.5 kg/kWe at MWe scale). **Verdict: a cost, not a gain.**



Figure 11: Verdict matrix (common closure, optimal-control trajectories). Colour intensity scales with the closure probability, and the small label gives SMAD's own verdict. Readiness, not closure, limits most cells.

Dropped. The options in Table 15 and the following results failed, and they are kept as negative results: hybrid chemical capture at Jupiter (initial mass +25 %, and +23 to +29 % with a chemical I_{sp} of 300–330 s); variable I_{sp} (an extra final mass at equal burn time of 3 kg at Mars and 110 kg at Jupiter, an upper bound, since the optimum constant I_{sp} is already near 4200 s); coasting instead of spiralling (0.1–2.2 % propellant saved for 11–123 % more time); perigee-only thrusting for Earth escape (not escaped after 1500 d at 10^{-3} m/s^2); thruster-cluster sizing beyond the power limit (0 %); lithium AF-MPD and hydrogen arcjet propellants (heavier or unclosable); and shielding only an electronics vault (no effect, because the half-angle is set by the radiator wings).

Reconcile-run results. (a) With the optimal-control solver, the propellant needed at a fixed trip time equals the burn-coast-burn value to within -0.9–0.7 % (Table 7). The loss of the family is in *time*, not propellant, because the impulsive plateau is reached; the minimum time is 3–46 % shorter (median 30 %) at the same thrust. (b) In the tier development plan, the converter life test is the critical path in both tiers. Accepting 50,000 h instead of 100,000 h moves the 2030s flight unit from 2044 to 2038, at the price of a 1.4–2.1× life gap on the outer-planet missions. This is a plan derived from assumed durations, not a forecast.

4.9 Sensitivity and "what would need to be true"

Single-assumption variants (this paper). Table 16 re-solves the 2030s minimum-mass designs with one assumption changed at a time. Halving the tank mass fraction saves 1 % of the Mars mass, about 5 % for Jupiter and 12 % for Uranus, so the propellant-heavy cases gain most. A 20 % lighter power system saves 6 % (Mars), 13 % (Jupiter) and 24 % (Uranus) of the initial mass. The capture definition is the strongest lever for Jupiter, where the power drops from 45 to 29 kWe, and it matters little for Uranus, where the heliocentric transit time dominates. The "minimum-mass" design is a flat optimum: a lighter power system pushes the best power *up* for Mars (71 against 64 kWe), because a faster trajectory then costs less mass.

What the verdicts hinge on (SMAD study, section 9.5). Table 17 lists the assumptions whose removal alone flips a verdict. In the 2030s era the verdicts depend on *evidence* (a 100,000-h converter test rather than 50,000 h, and critical items at TRL 6 rather than 4) and on *launch availability* (a Falcon-Heavy-only world loses the Mars and Jupiter closures). The mass assumptions change the launch mass by 13–58 %, but they do not flip a verdict by themselves except where the stack nears a vehicle limit. In the 2040s no single technology assumption flips a verdict (each costs 0–7 % of launch mass), but the verdicts are capped at "marginal" by TRL 5 evidence, and the only mission-enabling dependency is the refuelled-Starship capacity for the 20 t Mars cargo.

What would need to be true, in one list. (1) A flight-like reactor and shield ground test at power, and a deployed full-scale radiator (the 2030s milestones). (2) A converter life demonstration of at least the

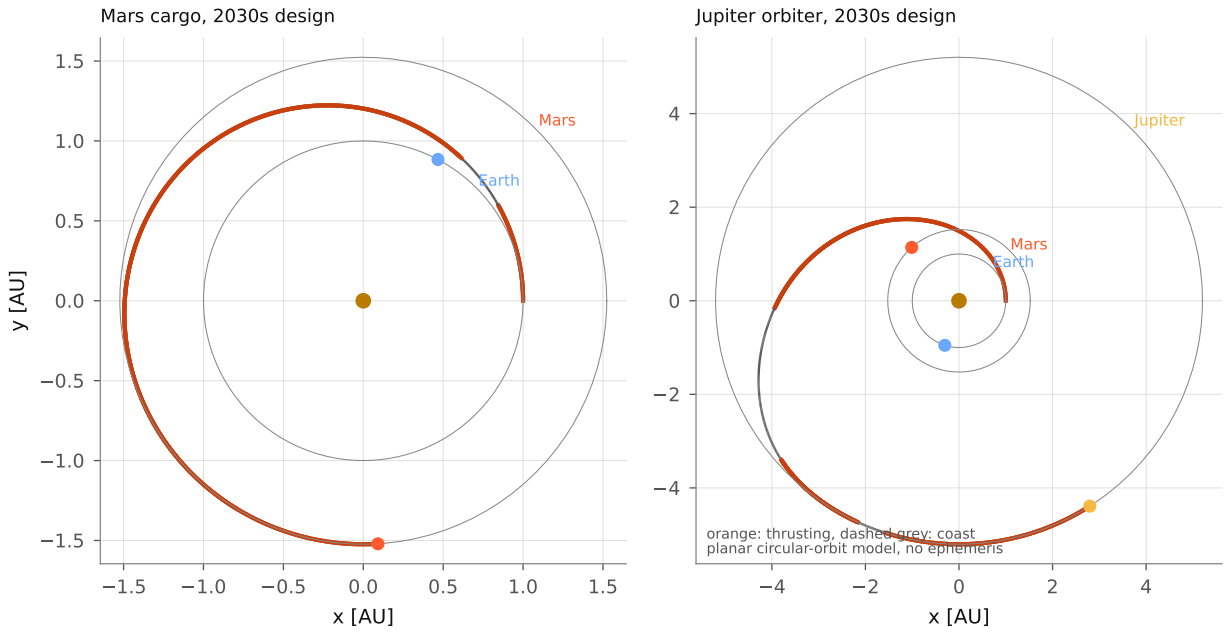


Figure 12: Optimal-control trajectories of the 2030s Mars and Jupiter designs in the planar model (orange: thrusting; dashed: coast). The capture spiral is not drawn. The Jupiter optimum coasts for a third of its heliocentric leg.

mission's full-power life (3, 8 and 12 yr for Mars, Jupiter and Uranus, i.e. 26,000, 70,000 and 105,000 h). (3) A launch vehicle for 15–25 t at $C_3 \approx 8$ (Mars) or 6–10 t at $C_3 = 40$ (Jupiter). (4) For the outer planets, a capture orbit and a transfer of the kind that the planar direct model of this paper over-penalises (gravity assists, lower capture orbits) or, failing that, power of order 100 kWe for Uranus. None of these is a physics limit; all are engineering and programmatic.

Uncertainty bands. The closure probabilities of Table 12 are 1.00 for most designs, so the informative spread is the sampled design power (5th–95th percentile in the table), which brackets the nominal value by roughly ±20–40 %. The fixed-hardware alternative (Table 18) is of order one half by construction. Neither includes structural model error (shield, Carnot fraction, bus scaling), which is larger.

5 Discussion

The results of Section 4 show that NEP is enabling where distance, not nominal power, is the problem, and that the mission verdicts are set by technology readiness rather than by mass closure. This section examines how far those findings can be trusted. It first lists the objections that a hostile reviewer would raise and then the numerical limits that remain.

5.1 What a reviewer would challenge

A hostile reading of this paper raises the following fifteen objections. Each was checked against the numbers, and the status says whether it was fixed, bounded or remains open.

1. **"The optimal-control trajectories are not proven optimal."** Open, bounded. A direct single-shooting search with 45 starts and a 20-node refinement returns feasible trajectories, and hence *upper* bounds on the minimum time and propellant. Re-integration with an independent integrator and convergence in the number of nodes were checked (Section 3), but no indirect-method (costate) certificate was computed. The real optimum can only be faster.
2. **"Planar, circular, coplanar planets with free phase ignore ephemerides, inclination and window waiting."** Open. The trip time counts from Earth escape to the capture orbit. The wait for a window (Mars synodic period 780 d), Earth escape from low orbit (7.7 km/s of ΔV if spiralled, excluded here by the stated C_3 launch) and the inclination (1.3–1.9°) are not costed. They add schedule and some ΔV, and a Lambert-based ephemeris model is the necessary next step.
3. **"The Jupiter verdict depends on how capture is defined."** Confirmed and quantified. With rendezvous plus a spiral to 30 planetary radii, the 2030s Jupiter orbiter needs 45 kWe; with a 1 km/s

Table 14: Supported design options from the SMAD study's paired Monte Carlo (near-term tier, N = 177, fixed seed; bands are lower bounds on the true spread) with the literature check. Gains are not additive. None removes a technology gate.

Idea	Change in launch mass, %: median [p10, p90] (i Mars, ii Jupiter, iii Uranus)	Cost / risk	Literature precedent (strict check)	Precedent	Verdict
Liquid-droplet radiator (LDR)	I -15.6 [-29.0, -8.0]; II -5.0 [-12.0, -0.8]; III -8.1 [-20.0, -1.8]	TRL 3; fluid loss, stream stability	Mattick & Hertzberg 1981 [24]; ground tests, not flown [22]	PRECEDENT (ground tests)	Keep
Higher-temperature Brayton (1500 K turbine inlet) with smaller radiator	I -11.6 [-26.5, -5.0]; II -2.5 [-6.5, -0.5]; III -4.0 [-11.8, -0.9]	TRL 3; new fuel form and alloys	Mason et al. 2021 [14]: 2.9 vs 1.6 MWe in the same radiator, "considerable development risk"	PRECEDENT	Keep
Deployable carbon-fibre heat-pipe fin radiator	I -14.7 [-26.2, -8.1]; II -7.5 [-14.2, -3.9]; III -11.1 [-21.8, -5.6]	TRL 4; loop and armour mass not demonstrated at 2.2 kg/m^2	NASA C-C radiator goal about 2 kg/m^2, no measured value [15]	PRECEDENT (goal)	Keep
Radiation-hardened payload/avionics (limits x4: 100 krad, 4e11 n/cm2)	I -4.6 [-9.0, -2.5]; II -3.0 [-5.0, -1.7]; III -4.1 [-8.3, -2.2]	neutron damage at 4e11 n/cm^2 limits optics and power parts	SMAD Table 8-7 [5]; no NEP-specific study opened	PARTIAL	Keep
Use the xenon tank as a gamma shield (tank between reactor and payload)	I -10.1 [-21.2, -5.3]; II -4.0 [-6.7, -2.3]; III -8.8 [-17.3, -4.7]	tank on axis in the umbra; credit shrinks as xenon is used	mercury propellant in the shadow shield of a NEP tug [23]; Xe and tanks credited qualitatively in [14]	PRECEDENT	Keep
Use PPU waste heat for bus heating (heaters off)	II -1.4 [-1.9, -1.2]; III -2.5 [-4.7, -2.0]	plumbing; heat only while thrusting	loop heat pipes cut NEXT PPU survival-heater power 183 to 35 W, not the same idea	PARTIAL	Keep (small)
Battery to shave payload power peaks	II -7.1 [-9.8, -6.0]; III -6.0 [-9.7, -5.0]	45 kg battery, cycling over 10 yr	standard practice for variable loads; no NEP source opened	NONE FOUND (routine)	Keep

Table 15: Options that failed the test (kept as negative results).

Idea	Change in launch mass, %	Result
Shield only an electronics vault (payload-zone radius 1.5 -> 0.6 m)	I +0.0 [+0.0, +0.0]; II +0.0 [+0.0, +0.0]; III +0.0 [+0.0, +0.0]	DROP
Lithium propellant at 7,000 s (AF-MPD) instead of xenon	I +144.0 [+72.1, +258.2]; II +18.6 [+3.6, +61.0]; III +12.4 [-7.1, +58.2]	DROP
Hydrogen arcjet at 1,000 s	I +77.7 [+44.3, +139.8]	DROP; DROP (does not close)

capture (the SMAD figure), the same closure gives 29 kWe. The truth lies between the two and depends on the science orbit and on whether an Oberth-assisted capture is feasible at low thrust, which it is not at these accelerations (E6 of the simulation: a single periapsis pass delivers metres per second).

4. **"You only unified the trajectory model, not the hardware."** True. The power-system mass tables are those of the SMAD study, which is conceptual: the shield model is conservative by a factor of 1.9 on mass against JIMO, the Brayton Carnot fraction is held constant with temperature, and the specific-mass fit is not valid above about 300 kWe. The agreement of the common closure with the SMAD native results (Table 8) shows consistency of method, not correctness of the hardware numbers. The only independent hardware check is the published NASA 20 kWe mass-equipment list [10].

5. **"The technology tiers are speculation."** The 2020s state is what exists or is funded, while the 2030s and 2040s states are conditional on milestones (Table 17, Fig. 10) and are labelled as such. Three of the four 2040s ingredients (1500 K conversion, 2.2 kg/m^2 radiators, 30,000-h high-I_{sp} thrusters) have no demonstration in the sources read.

6. **"Verdict thresholds are arbitrary."** Yes. The closure probabilities of 0.8 and 0.5, TRL 6 and 4, and life ratios of 1× and 2× are our judgement. The table gives the underlying numbers (Table 12), so a reader can apply another rule.

7. **"Closure probability is 1.00 almost everywhere, so it is uninformative."** Correct for the re-optimised definition: the uncertain inputs move the design power by tens of percent, but the launch capacity is not binding. The fixed-hardware probability (Table 18) is about one half by construction, because the design sits on the mass boundary. The informative quantity is the sampled power in Table 12.

8. **"Launch capacities are partly assumed."** The Falcon Heavy curve is a fit to one published point, SLS Block 1B availability is uncertain and the refuelled Starship curve is a placeholder, and the verdicts

Table 16: Single-assumption variants of the minimum-mass design. The capture definition is the largest lever for the outer planets.

Mission	Variant (optimal-control closure, 2030s)	P_e [kWe]	m_0 [t]	Δm_0 [%]
Mars cargo 8 t, 450 d	baseline (2030s, optimal control)	64	16.9	-
	tank mass fraction halved (staged tanks)	64	16.7	-1.1
	power-system mass −20 %	71	15.8	-6.2
Jupiter orbiter 1.5 t, 6 yr	baseline (2030s, optimal control)	45	9.4	-
	tank mass fraction halved (staged tanks)	43	9.0	-4.8
	power-system mass −20 %	43	8.2	-12.8
	capture ΔV 1 km/s (SMAD-like) instead of the spiral	29	6.6	-29.6
Uranus orbiter 1 t, 10 yr	baseline (2030s, optimal control)	105	17.9	-
	tank mass fraction halved (staged tanks)	94	15.8	-11.8
	power-system mass −20 %	89	13.5	-24.4
	capture ΔV 1 km/s (SMAD-like) instead of the spiral	92	14.9	-16.7

Table 17: SMAD study, section 9.5: the assumptions whose removal flips a verdict (all other single removals leave the verdict unchanged).

Era	Mission	ID	If this is NOT true	Base	Variant
2030s	(i) Mars cargo 8 t	A2	critical items stay at TRL 4 (no TRL-6 milestones)	Possible	Marginal
2030s	(i) Mars cargo 8 t	A3	power-system mass is the FSP-derived 142 kg/kWe curve, not the scaled reference	Possible	Marginal
2030s	(i) Mars cargo 8 t	A5	only Falcon Heavy available (no SLS, no Starship)	Possible	Does Not Close
2030s	(ii) Jupiter orbiter 1.5 t	A1	converter evidence only 50,000 h (not 100,000 h)	Possible	Marginal
2030s	(ii) Jupiter orbiter 1.5 t	A2	critical items stay at TRL 4 (no TRL-6 milestones)	Possible	Marginal
2030s	(ii) Jupiter orbiter 1.5 t	A5	only Falcon Heavy available (no SLS, no Starship)	Possible	Not Possible
2030s	(iii) Uranus orbiter 1 t	A1	converter evidence only 50,000 h (not 100,000 h)	Marginal	Not Possible
2030s	(iii) Uranus orbiter 1 t	A5	only Falcon Heavy available (no SLS, no Starship)	Marginal	Not Possible
2040s	(vi) Mars heavy cargo 20 t	B7	Starship 100 t instead of 150 t (no refuelling gain)	Marginal	Does Not Close

for the heavy Mars cases depend on the last two. Falcon-only availability flips the 2030s Mars and Jupiter verdicts (Table 17, rows A5).

9. **"Thrusters are modelled at constant I_{sp} and efficiency."** True. There are no throttle curves, and the NEXT hardware operated at 3270 s for Mars is taken from the SMAD study. The generic efficiency fit is used only between 1800 and 4300 s.

10. **"Power decay is ignored in the tables."** The optimal-control tables use constant power, but the closure applies the mean degradation factor to the thrust power (3.4×10^{-3}/yr, up to three times that in the Monte Carlo), which is a 1–3 % effect over 6 years.

11. **"There is no flight validation."** No NEP vehicle has flown. The nearest benchmark, Dawn, agrees to −7.5 % in xenon and −22 % in thrust hours (Section 3).

12. **"Bus mass is frozen at the SMAD design point."** True (errata E-S3). A ×0.9–1.15 multiplier is sampled, which brackets a 15 % error.

13. **"The 200-day Mars result depends on the paper's alpha convention."** It does not depend on the power-system model. The minimum time of 200 d needs an initial acceleration of 6.3e-04 m/s^2 and a propellant fraction above one half, so no mass model with $\alpha \gtrsim 10$ kg/kW closes. The lower bound on α for closure is given in `results/oc_gap.json`.

14. **"The novelty gains come from a small paired Monte Carlo and mostly lack primary sources."** Yes: $N = 177$, the bands are lower bounds and several precedents are abstract-only (Table 14). The table lists only what passed and flags the rest.

15. **"Cost, politics and nuclear approval are absent."** Cost is an analogy to JIMO (factor-2 confidence, $6–17 billion class in the SMAD study) and is not used in any verdict, and launch authorisation is a schedule item in the development plan. A verdict of "possible" therefore means technically closable, not affordable or approved.

Table 18: Alternative closure probability with the minimum-mass hardware frozen (power, thrusters, m_0) and inputs sampled: by construction about one half, because the design sits on the mass boundary; the columns show which constraint fails. The paper's verdicts use the re-optimised probability of Table 12.

Mission	Era	P(close), hardware fixed	fail: mass	fail: time	fail: life	fail: launch
Mars cargo 8 t, 450 d	2020s	0.21	0.55	0.52	0.00	0.00
	2030s	0.21	0.53	0.53	0.00	0.00
	2040s	0.26	0.74	0.00	0.00	0.00
Jupiter orbiter 1.5 t, 6 yr	2020s	0.21	0.64	0.04	0.39	0.00
	2030s	0.21	0.66	0.03	0.37	0.00
	2040s	0.19	0.73	0.00	0.28	0.00
	2030s	0.18	0.69	0.00	0.39	0.00
	2040s	0.18	0.75	0.00	0.27	0.00

5.2 Sources of error and model limitations

Numerical limits that remain. The tabulated propellant fractions carry solver noise of a few percent in the lowest-acceleration rows, where some nodes did not converge and were filled from neighbours (nodes filled per table: Mars isp3270 13, Mars isp4170 16, Jupiter 12, Uranus 43); designs near those rows are less certain than the tabulated values suggest. A direct optimal-control solve at the design points differs from the table by at most 9 % in propellant fraction (the table is the more conservative). Scaling the tabulated propellant fraction by ±5 % moves the minimum-mass design by under 1 % (Mars), about 2.5 % (Jupiter) and −11/+14 % (Uranus) in mass (`reconcile/selfcheck.py`). The Uranus rows are therefore the least certain numbers of the paper, and the minimum-time table for Uranus is bounded by the solver's time limit.

What was and was not re-run from scratch. The final pass of `run_all.py` rebuilt three of the five optimal-control tables from scratch (their cache keys had changed with a code revision) and ran every later stage (validation, gap study, reconciliation and Monte Carlo, animation data, figures, tables, this PDF) in sequence; `determinism_hashes.json` records its digests. Determinism evidence: (i) the Jupiter table and the Mars $C_3 = 0$ (2000 s) table rebuilt from scratch are bit-identical to their earlier builds (every minimum time and propellant fraction equal); (ii) the burn-coast-burn gap study (`oc_gap.json`, 22 direct optimal-control solves) re-ran to a byte-identical file; (iii) two re-solved cells of the Mars 4170 s table equal the stored values exactly (`reconcile/results/determinism_check.json`). The Mars 3270 s table (2020s and 2030s Mars designs) had first been built with an earlier start set (45 steering starts instead of the current 18); the rebuilt table differs from it by up to 3.5 % in minimum time (mean −0.4 %), and its least-converged propellant nodes differ by much more, which is the solver noise discussed above. The design points are not affected: the full reconciliation output (all designs, Monte Carlo distributions and variants) is identical to the previous run except two Mars escape masses that differ in the ninth significant digit. All numbers in this paper use the rebuilt tables. The Mars 4170 s and Uranus tables were not rebuilt in the final pass.

6 Conclusion

This work audited, reproduced and rebuilt the 2025 analysis of nuclear electric propulsion, reconciling two independent models on one documented input set and checking their trajectories against a new optimal-control solver.

At the level of the original study, the headline numbers do not follow from its own equations: Table 1 is off by 0.27 for NEP and by 0.53 for the other rows, the constant-mass propellant exceeds the spacecraft mass, the asymptotes of Fig. 8 lie at 112 and 130 d rather than 150 and 250 d, and the ten-RTG mission can never complete. Once corrected, NEP is about 10× faster than solar at the same thruster power, and it is decisively better only beyond 2–3 AU at equal power-system mass. At Mars, a solar-electric craft of equal power-system mass is only about 8 % slower (393 against 362 d).

At the level of the models, the simulation and the SMAD study are reconcilable for Mars, to 5–15 % in power and mass, once the simulation's failed trajectory-table nodes (35 of 63 for Mars at $C_3 = 8.2$) are removed. Its "closes in every tier" result was an artefact, and its Mars 200-day result is withdrawn. The two models are *not* reconcilable for Jupiter and Uranus without choosing a capture definition and a transfer type, which differ between them by a factor of 2–6 in power. The burn-coast-burn family is a safe upper bound: the optimal-control solver finds the same propellant at equal time and trip times 3–46 % shorter (median 30 %), with the gap growing with thrust acceleration. This bound is not proven optimal.

At the mission level, on common inputs, a Mars cargo of 8 t in 450 d needs about 95, 64 and 80 kWe in the 2020s, 2030s and 2040s (it is power-limited, not mass-limited), and a 1.5 t Jupiter orbiter needs 89,

45 and 40 kWe. The mission verdicts are set by technology readiness (reactor, converter life and radiator at TRL 4–5 today) rather than by physics, and Table 12 gives them. The 2030s state needs a 100,000-h converter test whose critical path ends in 2044 (2038 if 50,000 h is accepted), and the 2040s state ends in 2050. Of the design ideas, the supported ones are the xenon-tank gamma credit, the carbon-fibre radiator, radiation-hardened avionics, PPU-heat reuse and tank staging for long missions, while the liquid-droplet radiator and the 1500 K Brayton are TRL 3 research options. All but PPU-heat reuse and the battery have literature precedent, so the contribution of this paper is the quantification, not the ideas.

In general, the results demonstrate that the advantage of NEP is real but comes from distance rather than from the nominal power factor, and that whether a mission is possible is decided by technology readiness rather than by mass closure. The original study's claims are therefore not supported in the form or by the margins in which they were stated. The main open uncertainties are the capture and transfer definition for the outer planets, the conceptual hardware masses (shield, conversion), the launch-vehicle curves for the heavy cases, and the unproven global optimality of the trajectories. Future work should address these uncertainties, beginning with the Lambert-based ephemeris model identified in Section 5.

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A Reproducibility Statement

Code and data. All inputs are kept in `NEP_rebuild/data` (parameters with source and status), `sim` (trajectory simulation), `smad` (SMAD design study) and `audit`, and this paper adds only files under `NEP_rebuild/paper`. The command `python paper/run_all.py` (plain `python`; numpy, scipy, matplotlib, numba; xelatex) regenerates the optimal-control tables (`reconcile/build_tables.py`), the solver validation, the burn-coast-burn gap study, the reconciliation with its Monte Carlo (seed 20251103, Latin hypercube), the animation data, all figures in the paper and the dark-theme sets, all tables and macros, and this PDF. No number in the text is typed by hand. With `--clean`, the tables are rebuilt from scratch.

Determinism. The solver has no random element (fixed start list, SLSQP, fixed-step RK4), and the only use of random numbers is the seeded Latin hypercube. The file `paper/determinism_hashes.json` records SHA-256 digests of the results files (and of their numerical content rounded to nine significant digits), of every optimal-control table array and of the inputs, and a clean rebuild must reproduce them (Section 5 states what was and was not re-run).

Original packages untouched. The `sim` and `smad` packages are imported read-only, and the sim trajectory cache is read from a copy in `reconcile/cache_bcb`. All defects found in them are documented in `reconcile/errata.md` and corrected only inside the reconcile code.

Sources actually read. Only sources whose full text, abstract page or specific tables were opened are cited, and the reference list marks abstract-only items. The following were not opened and are therefore not cited as evidence: the original Irving and Blum (1959) report, Goebel and Katz, Edelbaum (1961), Kluever, Betts and Conway on low-thrust optimisation, and the Purdue work on NEP gravity assists (access was blocked). Standard results attributed to them (the Stuhlinger optimum and the tangential-spiral $\Delta V = |v_{c1} - v_{c2}|$) are instead checked numerically in Section 3 or sourced to the lecture notes and abstract pages listed. Values recalled without a document (SNAP-10A, Europa Clipper mass, xenon production) are marked as such in the SMAD study and are not load-bearing for any verdict here.

Statuses used throughout. *Computed* denotes the output of one of the models, *literature* a value read in a cited document, and *assumed* a stated assumption with a range. The technology tiers (2020s, 2030s, 2040s) are technology *states* with stated milestones, not forecasts of dates.

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